ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Current Electricity

English Medium

EXERCISE # O-1

SINGLE CORRECT TYPE QUESTIONS

A. Microscopic Ohm's Law

1. Two wires each of radius of cross section r but of different materials are connected together end to end (in series). If the densities of charge carries in the two wires are in the ratio 1 : 4, the drift velocity of electrons in the two wires will be in the ratio :

(A) 1 : 2 (B) 2 : 1 (C) 4 : 1 (D) 1 : 4

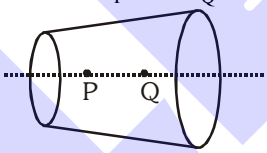
CE0251

2. A current I flows through a uniform wire of diameter d when the mean electron drift velocity is v. The same current will flow through a wire of diameter d/2 made of the same material if the mean drift velocity of the electron is :

(A) v/4 (B) v/2 (C) 2v (D) 4v

CE0252

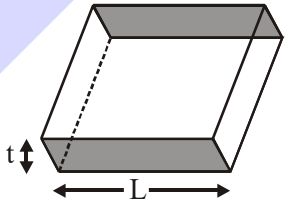
3. A wire has a non–uniform cross–section as shown in figure. A steady current flows through it. The drift speed of electrons at points P and Q is v_P and v_Q :



(A) $v_P = v_Q$ (B) $v_P < v_Q$ (C) $v_P > v_Q$ (D) data insufficient

CE0253

4. Consider a thin square sheet of side L and thickness t, made of a material of resistivity ρ. The resistance between two opposite faces, shown by the shaded areas in the adjoining figure is :-



(A) directly proportional to L (B) directly proportional to t

(C) independent of L (D) independent of t

CE0254

5. If the current in resistance of uniform cross section is doubled then choose **INCORRECT** option (assume temperature to be approximately constant) :-

(A) the current density is doubled

(B) the conduction electron density is doubled

(C) the mean time between collision is constant

(D) the electron drift speed is doubled

CE0255

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6. The resistance of a wire is 1 ohm. The wire is stretched to double its length. Now, resistance of the wire is

(A) $\frac{1}{4}$ ohm (B) 4 ohm (C) 2 ohm (D) 8 ohm

CE0256

7. If a copper wire is stretched to make it 0.1% longer. Then the percentage change in resistance is approximately.

(A) 0.1 % (B) 0.2 % (C) 0.4 % (D) 0.8 %

CE0257

8. Electromotive force represents

(A) Force (B) Momentum

(C) work (D) energy per unit charge

CE0258

9. The temperature-coefficient of resistance of a wire is 0.00125 per °C. Its resistance at 300K is 1 ohm. Its resistance will be 2 ohm at

(A) 1154 K (B) 1100 K (C) 1400 K (D) 1127 K

CE0259

10. If n, e, τ and m represent electron density, charge, average relaxation time and mass respectively, then the resistance of a wire of length l and cross-section area A is given by

(A) $\frac{ml}{ne^2\tau A}$ (B) $\frac{m\tau A}{ne^2 l}$ (C) $\frac{ne^2\tau A}{2ml}$ (D) $\frac{ne^2 A}{2m\tau l}$

CE0260

11. The resistance of a conductor is 5 ohm at 50°C and 6 Ω At 100°C. What is its resistance at 0°C ?

(A) 1 ohm (B) 2 ohm (C) 3 ohm (D) 4 ohm

CE0261

12. Which of the following characteristic of electrons determines the current in a conductor ?

(A) Drift velocity alone

(B) Thermal velocity alone

(C) Both drift and thermal velocity

(D) Neither drift nor thermal velocity

CE0262

13. The conductivity of a conductor decreases, with temperature because on heating

(A) atoms vibrate vigorously (B) conductor expands

(C) electrons gain energy (D) electrons vibrate vigorously

CE0263

14. The resistance of a wire of iron is 10 ohm and temperature coefficient of resistivity is $5\times10^{-3}/°C$. At 20°C it carries 30 mA of current. Keeping constant potential difference between its ends, the temperature of the wire is raised to 120°C. The current in milliamperes that flows in the wire now is

(A) 20 (B) 15 (C) 10 (D) 40

CE0264

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15. Two conductors have the same resistance at 0°C but their temperature coefficients of resistance are α_1 and α_2. The respective temperature coefficients of their series and parallel combinations are nearly :

[AIEEE-2010]

(A) $\frac{\alpha_1+\alpha_2}{2}, \frac{\alpha_1+\alpha_2}{2}$ (B) $\frac{\alpha_1+\alpha_2}{2}, \alpha_1+\alpha_2$

(C) $\alpha_1+\alpha_2, \frac{\alpha_1+\alpha_2}{2}$ (D) $\alpha_1+\alpha_2, \frac{\alpha_1\alpha_2}{\alpha_1+\alpha_2}$

CE0131

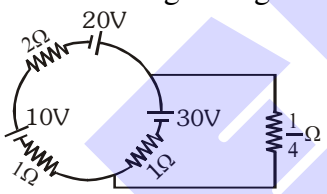
B. Kirchoff's Laws

16. Potential difference of a cell in an open circuit is 6 volts but it falls to 4 volts when a current of 2 amp is drawn from it. The internal resistance of the cell is

(A) 1Ω (B) 2Ω (C) 3Ω (D) 4Ω

CE0265

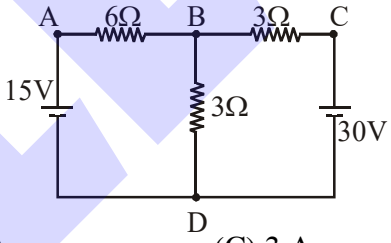
17. In the given circuit diagram, the current flowing through resistor of 1/4 Ω is :



(A) 1 A (B) 60 A (C) 30 A (D) None of these

CE0266

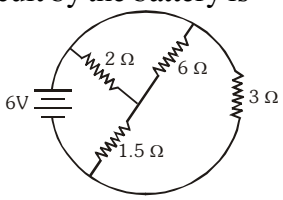
18. In the circuit shown in figure, find the current through the branch BD :



(A) 5 A (B) 0 A (C) 3 A (D) 4 A

CE0267

19. The total current supplied to the circuit by the battery is-



(A) 1 A (B) 2 A (C) 4 A (D) 6 A

CE0268

20. A primary cell has an e.m.f 1.5 volt. When short-circuited, it gives a current of 3 amp. The internal resistance of the cell is

(A) 4.5 Ω (B) 2 Ω (C) 0.5 Ω (D) 1/4.5 Ω

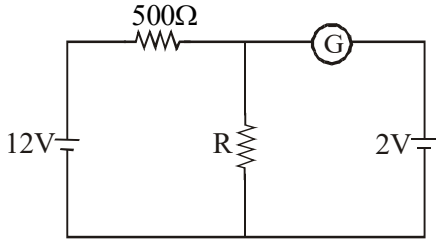
CE0269

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21. The resistance of flash light is 1.5Ω. It uses 3 cells each of 1.5 V and internal resistance 0.25Ω. What is the net potential difference across the lamp ?

(A) 1.5 V (B) 2.25 V (C) 3 V (D) 4.5 V **CE0270**

22. In the given circuit the galvanometer reads zero. The value of resistance R is



(A) 50Ω (B) 100Ω (C) 200Ω (D) 500Ω

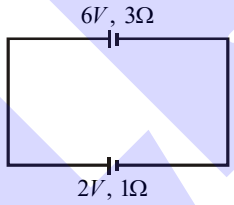
CE0271

23. When a resistance of 9.5 ohm is connected across a battery, the voltage across the resistance is 11.4 V. If the resistance connected across the same battery is 11.5 ohm, the voltage across the resistance is 11.5 V

(A) The emf of the battery is 12.0 V
(B) The internal resistance of the battery is 0.6 ohm
(C) The emf of the battery is 11.45 V
(D) The emf of the battery is 11.5 V

CE0272

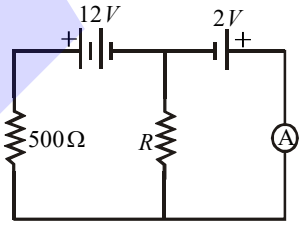
24. What the current in the circuit given below ?



(A) 4 A (B) 2 A (C) 1 A (D) 0.5 A

CE0273

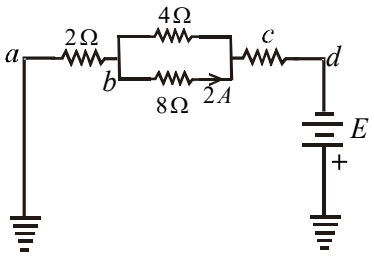
25. In the circuit given below, the ammeter reading is zero. What is the value of the resistance R?



(A) 50Ω (B) 100Ω (C) 200Ω (D) 400Ω

CE0274

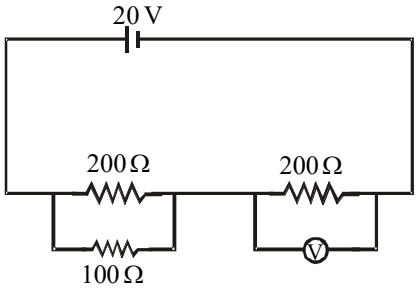
26. In the circuit given below, what is the current in the 2Ω resistor ?



(A) 2 A (B) 4 A (C) 6 A (D) 8 A **CE0275**

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27. In the following circuit, what is the reading of the voltmeter ?



(A) 5 V (B) 10 V (C) 15 V (D) 16.5 V

CE0276

28. In the above question, if the reading of voltmeter is 10V, then resistance of the voltmeter is

(A) 50Ω (B) 100Ω (C) 200Ω (D) 300Ω

CE0277

C. Equivalent Resistance and equivalent of battery

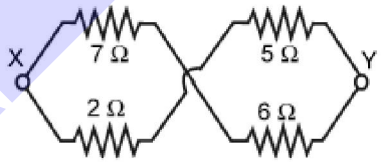
29. G_1, G_2, G_3 are the conductances of the three conductors. When they are joined in series, their equivalent conductance will be

(A) $G_1 + G_2 + G_3$ (B) $\frac{1}{G_1}+\frac{1}{G_2}+\frac{1}{G_3}$

(C) $\frac{G_1 G_2 G_3}{G_1+G_2+G_3}$ (D) given by some other relation

CE0278

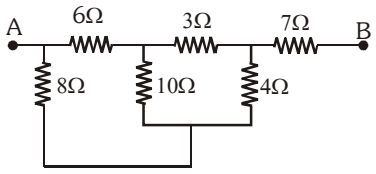
30. Net resitance between X and Y is



(A) 4 Ω (B) 4.55 Ω (C) 2 Ω (D) 20 Ω

CE0279

31. The equivalent resistance between the points A and B is :

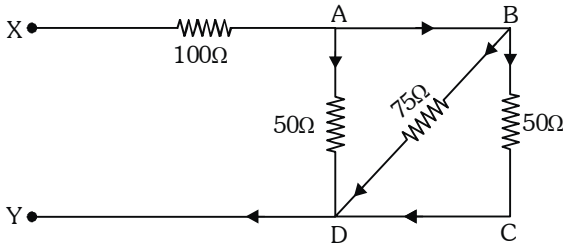


(A) $\frac{36}{7}\Omega$ (B) 10 Ω (C) $\frac{85}{7}\Omega$ (D) none of these

CE0280

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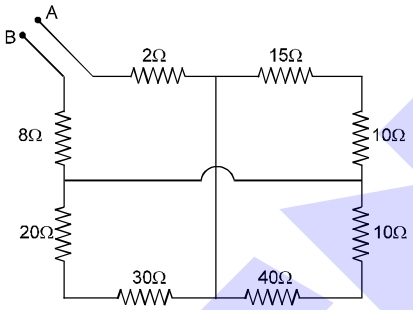
32. In an electrical arrangement as shown the equivalent resistance between X and Y will be:



(A) 150/8 Ω (B) 750/8 Ω (C) 950/8 Ω (D) 850/8 Ω

CE0281

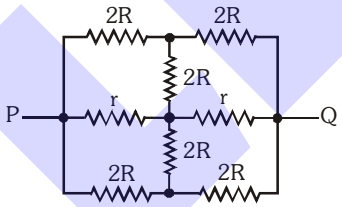
33. The equivalent resistance between points A and B is :



(A) 32.5 Ω (B) 22.5 Ω (C) 2.5 Ω (D) 45.5 Ω

CE0282

34. The effective resistance between points P and Q of the electrical circuit shown in the figure is :



(A) $\frac{2Rr}{R+r}$ (B) $\frac{8R(R+r)}{3R+r}$ (C) 2r + 4R (D) $\frac{5R}{2}+2r$

CE0283

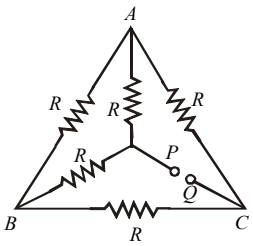
35. A technician has only two resistors. By using them singly, in series or in parallel, he is able to obtain the resistances of values 3, 4,12 and 16 Ω. What are the resistances of the two coils ?

(A) 6Ω *and* 10Ω (B) 4 *and* 12Ω

(C) 9 *and* 9Ω (D) 5 *and* 12Ω

CE0284

36. Each resistance in the network is R. The equivalent resistance between points P and Q is

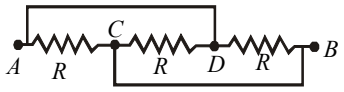


(A) R (B) 5R (C) 3R (D) 6R

CE0285

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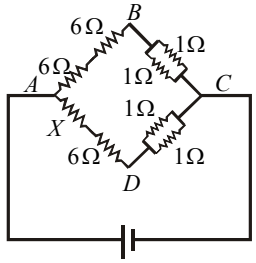
37. Find the resistance of the circuit between the points A and B



(A) R/3 (B) 2R/3 (C) 3R/2 (D) 3R

CE0286

38. For what value of the unknown resistance X, the potential difference between B and D will be zero in the arrangement of adjoining figure.



(A) 4Ω (B) 2Ω (C) 3Ω (D) 6Ω

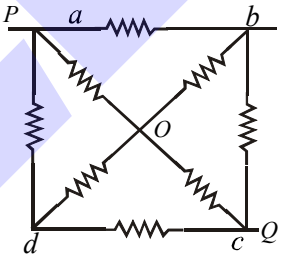
CE0287

39. Four wire AB,BC,CD and DA, each of resistance 4 ohm, and a fifth wire BD of resistance 8 ohm are joined to form a rectangle of which BD is a diagonal. The effective resistance between the points A and B is

(A) 24 Ω (B) 16 Ω (C) 4/3 Ω (D) 8/3 Ω

CE0288

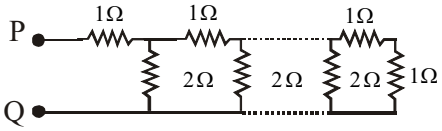
40. What is resistance between P & Q in the following network. Each resistance is 12Ω.



(A) 24Ω (B) 16Ω (C) 8Ω (D) 4Ω

CE0289

41. What is resistance between P & Q in the following infinite ladder net work of resistances ?

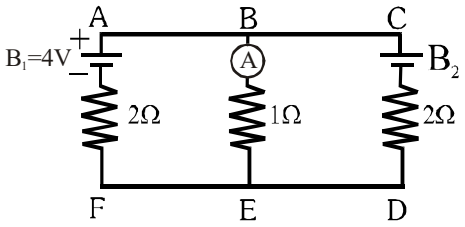


(A) ∞ (B) 4Ω (C) 3Ω (D) 2Ω

CE0290

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42. In the diagram shown, what will be the value emf (in volt) of the battery B_2, if the ammeter reads zero current–



(A)2 (B) 3 (C) 4 (D) zero

CE0291

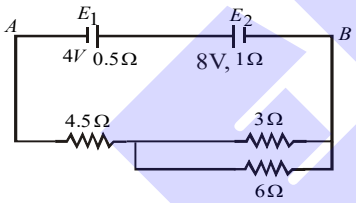
43. Five cells, each of e.m.f E and internal resistance r are connected in series. If due to over sight, one cell is connected wrongly, then the equivalent e.m.f and internal resistance of the combination, is

(A) 5E and 5r (B) 3E and 3r

(C) 3E and 5r (D) 5E and 3r

CE0292

44. In the circuit of figure, potential difference across E_1 and E_2 will be



(A) $4.25V, 4.25V$ (B) $3.75V, 3.75V$

(C) $4.25V, 7.5V$ (D) $3.75V, 7.5V$

CE0293

45. Five cells each of internal resistance $0.2\ \Omega$ and e.m.f 2 V are connected in series with a resistance 4Ω. The current through the external resistance is

(A) 0.2 A (B) 00.5 A (C) 1 A (D) 2 A

CE0294

46. A battery of 6 V and internal resistance 2Ω sends a current of 1 A through a load. If two such batteries are connected in series across the same load, the current through the load will be

(A) 1.20 A (B) 1.50 A

(C) 2.00 A (D) none of these

CE0295

47. You are given a wire 1m long. In which one of the following cases, the energy drawn from the battery will be the largest ? Assume that the internal resistance of the battery is zero.

(A) Full length of the wire connected across the battery

(B) Half length of the wire connected across the battery

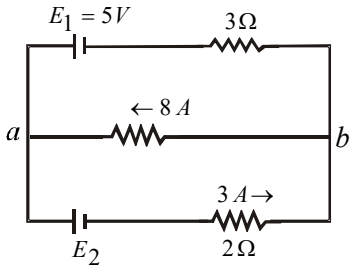
(C) Wire cut into four equal parts and they are connected in parallel to each other across the battery.

(D) Wire is cut into two equal parts and they are connected in parallel to each other across the battery.

CE0296

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48. What is the current through the 3Ω resistance in the circuit shown below ?



(A) 5 A (B) 3 A

(C) 11 A (D) cannot be calculated unless E_2 is known

CE0297

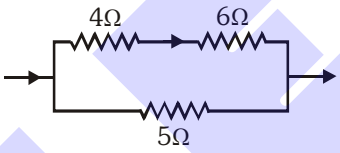
49. If $400\,\Omega$ of resistance is made by adding four $100\,\Omega$ resistance of tolerance 5%, then the tolerance of the combination is : **[AIEEE - 2011]**

(A) 20% (B) 5% (C) 10% (D) 15%

CE0133

D. Heating effect due to current and Electrical Instruments

50. In the circuit shown in figure the heat produced in the 5Ω resistor due to the current flowing through it is 10 cal/s. The heat generated in the 4Ω resistor is :



(A) 1 cal/s (B) 2 cal/s (C) 3 cal/s (D) 4 cal/s

CE0298

51. What is the maximum number of 100W, 200V lamps which can be connected in a circuit having fuse wire of safe current 5A ?

(A) 5 (B) 10 (C) 20 (D) 40

CE0299

52. The safe current of a fuse wire does not depend on its

(A) length (B) radius

(C) area of cross-section (D) material of the wire

CE0300

53. If the conductivity of the fuse wire is doubled, the safe current will be

(A) 2 times (B) $\sqrt{2}$ times

(C) 1/2 times (D) $1/\sqrt{2}$ times

CE0301

54. Two heater wires of same length and material but of different thickness are connected in series across a power supply. The power dissipated

(A) will be same in both

(B) will be more in thinner wire

(C) will be more in thicker wire

(D) cannot be predicted

CE0302

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55. The resistance of hot tungsten filament is about 10 times the cold resistance. What will be the resistance of 100W and 200 V lamp when not in use ?

(A) 14000Ω (B) 400Ω (C) 40Ω (D) 4Ω

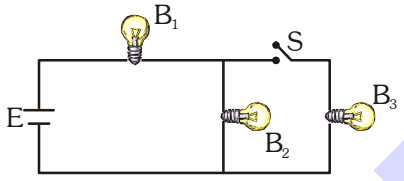
CE0303

56. Just as electricity is supplied at 220 V for domestic use in India, it is supplied at 110 V in USA. If resistance of 60 W bulb for use in India is R, that of a 60 W bulb for use in USA will be

(A) R/4 (B) R/2 (C) R (D) 2R

CE0304

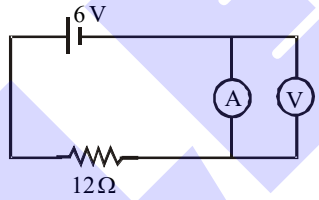
57. Three identical bulbs are connected as shown in figure. When switch S is closed, the power consumed in bulb B_2 is P. The power consumed by same bulb when switch is open in–



(A) $\frac{4P}{9}$ (B) $\frac{9P}{4}$ (C) $\frac{9P}{16}$ (D) $\frac{16P}{9}$

CE0305

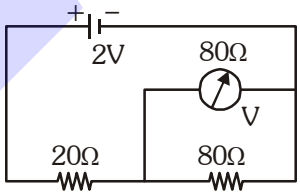
58. In the circuit given below, the ammeter and voltmeter are ideal measuring devices. What is the reading of the voltmeter ?



(A) 6 V (B) 3 V (C) 1.5 V (D) zero

CE0306

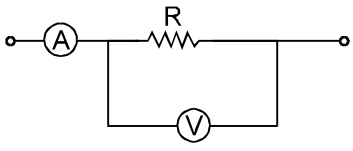
59. The reading of voltmeter in the following circuit figure will be–



(A) 2 volt (B) 0.80 volt (C) 1.33 volt (D) 1.60 volt

CE0307

60. In the circuit shown the readings of ammeter and voltmeter are 4A and 20V respectively. The meters are non-ideal, then R is



(A) 5 Ω (B) less than 5Ω
(C) greater than 5Ω (D) between 4Ω and 5Ω.

CE0308

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grow\Current Elec.-Copy\Eng\03_Current Electricity\01_Eng.p65

61. A galvanometer of resistance 25Ω gives full-scale deflection for a current of 10mA. It is to be changed into a voltmeter of range 100 V by connecting a resistance R in series with the galvanometer coil. The value of R in ohms is

(A) 10, 000 (B) 10,025 (C) 975 (D) 9975

CE0309

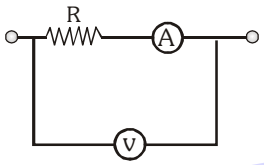
62. A galvanometer has a resistance of 20Ω and reads full–scale when 0.2V is applied across it. To convert it into a 10A ammeter, the galvanometer coil should have a

(A) 0.01 Ω resistor connected across it (B) 0.02 Ω resistor connected across it

(C) 200 Ω resistor connected in series with it (D) 2000 Ω resistor connected in series with it

CE0310

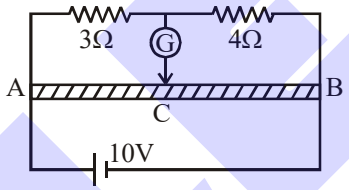
63. In the circuit shown the resistance of voltmeter is 10,000 ohm and that of ammeter is 20 ohm. The ammeter reading is 0.10 Amp and voltmeter reading is 12 volt. Then R is equal to



(A) 122 Ω (B) 140Ω (C)116 Ω (D) 100Ω

CE0311

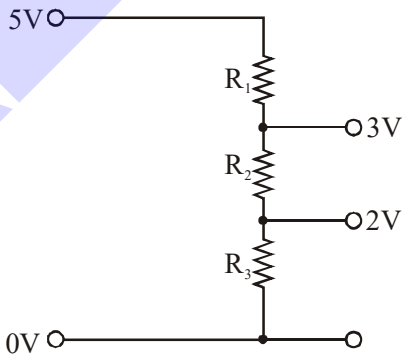
64. The galvanometer deflects zero, and the length of wire AB is 350 cm. then AC equals to :-



(A) 150 cm. (B) 200 cm. (C) 210 cm. (D) 280 cm.

CE0312

65. A potential divider is used to give outputs of 2V and 3V from a 5V source, as shown in figure. Which combination of resistances, R_1, R_2 and R_3 gives the correct voltages?



	R_1	R_2	R_3
(A)	1 kΩ	1 kΩ	2 kΩ
(B)	2 kΩ	1 kΩ	2 kΩ
(C)	3 kΩ	2 kΩ	2 kΩ
(D)	3 kΩ	2 kΩ	3 kΩ

CE0313

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Copo\Eng\03_Current Electricity\01_Eng.p65

66. The length of a potentiometer wire is ℓ. A cell of emf E is balanced at a length $\ell/3$ from the positive end of the wire. If the length of the wire is increased by $\ell/2$ at what distance will the same cell give a balanced point (assuming battery in primary circuit is ideal)

(A) $\frac{2\ell}{3}$ (B) $\frac{\ell}{2}$ (C) $\frac{\ell}{6}$ (D) $\frac{4\ell}{3}$

CE0314

67. For higher sensitivity which of the following is essential for the potentiometer ?

(A) higher emf of the auxiliary battery

(B) higher specific resistance of wire

(C) larger length of wire

(D) none of the above

CE0315

68. A 4m long wire of resistance 8Ω is connected in series with a battery of emf 2V and a resistor of 7Ω. The internal resistance of the battery is 1Ω. What is the potential gradient along the wire ?

(A) $0.25 Vm^{-1}$ (B) $0.50 Vm^{-1}$ (C) $0.75 Vm^{-1}$ (D) $1.00 Vm^{-1}$

CE0316

69. The current in the primary circuit of a potentiometer is 0.2 A. The specific resistance and cross-section of the potentiometer wire are 4×10^{-7} ohm metre and 8×10^{-7} m^2 respectively. The potential gradient will be equal to :- **[AIEEE - 2011]**

(A) 0.2 V/m (B) 1 V/m (C) 0.5 V/m (D) 0.1 V/m

CE0134

70. A resistor has a color code of green, blue; brown and silver. What is its resistance?

(A) 56 Ω ± 5% (B) 560 Ω ± 10%

(C) 560 Ω ± 5% (D) 5600 Ω ± 10%

CE0106

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\Current Elec.-Copy\Eng\03_Current Electricity\01_Eng.p65

EXERCISE # O-2

SINGLE CORRECT TYPE QUESTIONS

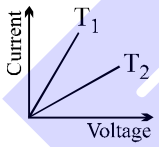
A. Microscopic Ohm's Law

1. An insulating pipe of cross-section area 'A' contains an electrolyte which has two types of ions→ their charges being –e and +2e. A potential difference applied between the ends of the pipe result in the drifting of the two types of ions, having drift speed = v (–ve ion) and v/4 (+ve ion). Both ions have the same number per unit volume = n. The current flowing through the pipe is

(A) nev A/2 (B) nev A/4 (C) 5nev A/2 (D) 3nev A/2

CE0044

2. The current in a metallic conductor is plotted against voltage at two different temperatures T_1 and T_2. Which is correct



(A) $T_1 > T_2$ (B) $T_1 < T_2$ (C) $T_1 = T_2$ (D) none

CE0045

3. A metal rod of length 10 cm and a rectangular cross-section of 1 cm $\times \frac{1}{2}$ cm is connected to a battery across opposite faces. The resistance will be

(A) maximum when the battery is connected across 1 cm $\times \frac{1}{2}$ cm faces.

(B) maximum when the battery is connected across 10 cm × 1 cm faces.

(C) maximum when the battery is connected across 10 cm $\times \frac{1}{2}$ cm faces.

(D) same irrespective of the three faces.

CE0046

4. Consider a current carrying wire (current I) in the shape of a circle. Note that as the current progresses along the wire, the direction of **j** (current density) changes in an exact manner, while the current I remain unaffected. The agent that is *essentially* responsible for is

(A) source of emf.

(B) electric field produced by charges accumulated on the surface of wire.

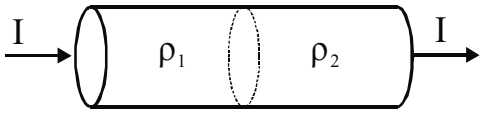
(C) the charges just behind a given segment of wire which push them just the right way by repulsion.

(D) the charges ahead.

CE0047

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Copy\Eng\03_Current Electricity\01_Eng.p65

5. Two long straight cylindrical conductors with resistivities ρ_1 and ρ_2 respectively are joined together as shown in figure. If current I flows through the conductors, the magnitude of the total free charge at the interface of the two conductors is :-



(A) zero (B) $\frac{(\rho_1-\rho_2)I\varepsilon_0}{2}$ (C) $\varepsilon_0 I|\rho_1-\rho_2|$ (D) $\varepsilon_0 I|\rho_1+\rho_2|$

CE0048

6. **Statement 1 :** The drift speed of electrons in metals is small (in the order of a few mm/s) and the charge of an electron is also very small (= 1.6×10^{-19}C), yet we can obtain a large current in a metal.
and
Statement 2: At room temperature, the thermal speed of electron is very high (about 10^7 times the drift speed).
(A) Statement–1 is True, Statement–2 is True ; Statement–2 is a correct explanation for Statement–1.
(B) Statement–1 is True, Statement–2 is True ; Statement–2 is not a correct explanation for Statement–1.
(C) Statement–1 is True, Statement–2 is False.
(D) Statement–1 is False, Statement–2 is True.

CE0049

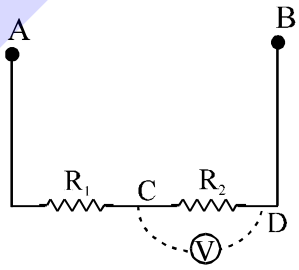
7. Two resistances of equal magnitude R and having temperature coefficient α_1 and α_2 respectively are connected in parallel. The temperature coefficient of the parallel combination is, approximately

(A) $2(\alpha_1+\alpha_2)$ (B) $\frac{\alpha_1\alpha_2}{\alpha_1+\alpha_2}$ (C) $\frac{\alpha_1-\alpha_2}{2}$ (D) $\frac{\alpha_1+\alpha_2}{2}$

CE0058

B. Kirchoff's Laws

8. Resistances R_1 and R_2 each 60Ω are connected in series as shown in figure. The Potential difference between A and B is kept 120 volt. Then what will be the reading of voltmeter connected between the point C & D if resistance of voltmeter is 120Ω.



(A) 48 V (B) 24 V (C) 40V (D) None

CE0052

9. A simple circuit contains an ideal battery and a resistance R. If a second resistor is placed in parallel with the first,
(A) the potential across R will decrease
(B) the current through R will decreased
(C) the current delivered by the battery will increase
(D) the power dissipated by R will increased. **CE0054**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grow\3-Current Elec.-Copa\Eng\03_Current Electricity\01_Eng.p65

10. An energy source will supply a constant current into the load, if its internal resistance is-

[AIEEE - 2005]

(A) equal to the resistance of the load
(B) very large as compared to the load resistance
(C) zero
(D) non-zero but less than the resistance of the load

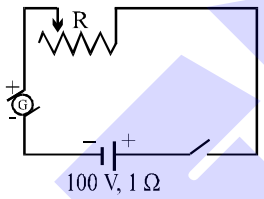
CE0056

11. When a current of 4 A flows within a battery from its positive to negative terminal, the potential difference across the battery is 12 volts. The potential difference across the battery is 9 volts when a current of 2 A flows within it from its negative to its positive terminal. The internal resistance and the e.m.f. of the battery are :-

(A) 0.1 Ω, 4V (B) 0.2 Ω, 5V (C) 0.5 Ω, 10V (D) 0.7 Ω, 10V

CE0057

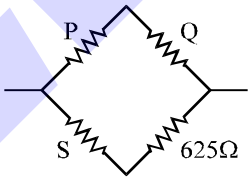
12. The battery in the diagram is to be charged by the generator G. The generator has a terminal voltage of 120 volts when the charging current is10 amperes. The battery has an emf of 100 volts and an internal resistance of 1 ohm. In order to charge the battery at 10 amperes charging current, the resistance R should be set at :-



(A) 0.1 Ω (B) 0.5 Ω (C) 1.0 Ω (D) 5.0 Ω

CE0063

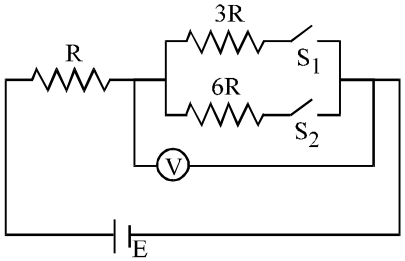
13. A Wheatstone's bridge is balanced with a resistance of 625 Ω in the third arm, where P, Q and S are in the 1st, 2nd and 4th arm respectively. If P and Q are interchanged, the resistance in the third arm has to be increased by 51Ω to secure balance. The unknown resistance in the fourth arm is :-



(A) 625 Ω (B) 650 Ω (C) 676 Ω (D) 600 Ω

CE0065

14. In the circuit shown in figure reading of voltmeter is V_1 when only S_1 is closed, reading of voltmeter is V_2 when only S_2 is closed. The reading of voltmeter is V_3 when both S_1 and S_2 are closed then

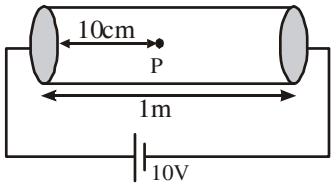


(A) $V_2 > V_1 > V_3$ (B) $V_3 > V_2 > V_1$ (C) $V_3 > V_1 > V_2$ (D) $V_1 > V_2 > V_3$

CE0067

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Capa\Eng\03_Current Electricity\01_Eng.p65

15. A cylindrical solid of length 1m and radius 1 m is connected across a source of emf 10V and negligible internal resistance shown in figure. The resistivity of the rod as a function of x (x measured from left end) is given by $\rho = bx$ [where b is a positive constant]. Find the electric field (in SI unit) at point P at a distance 10 cm from left end.



(A) 1 (B) 2 (C) 3 (D) 4

CE0071

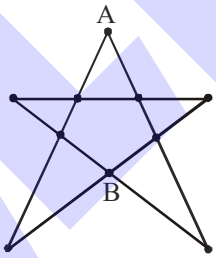
16. An electric bell has a resistance of 5Ω and requires a current of 0.25 A to work it. Assuming that the resistance of the bell wire is 1Ω per 15m and that the bell push is 90m distance from the bell. How many cells each of emf 1.4V and internal resistance 2Ω, will be required to work the circuit-

(A) 3 (B) 4 (C) 5 (D) Can't be determined

CE0074

C. Equivalent Resistance and equivalent of battery

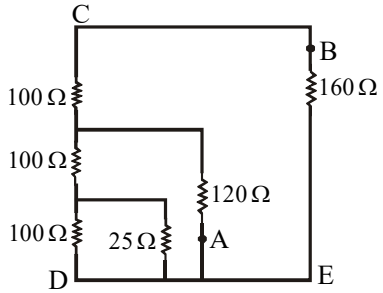
17. The resistance of all the wires between any two adjacent dots is R. The equivalent resistance between A and B as shown in figure is



(A) $\frac{7}{3}$ R (B) $\frac{7}{6}$ R (C) $\frac{14}{8}$ R (D) None of these

CE0317

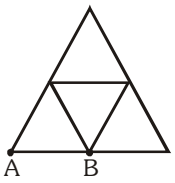
18. What is the equivalent resistance across the points A & B in the circuit given below



(A) 80Ω (B) 120Ω (C) 160Ω (D) 320Ω **CE0318**

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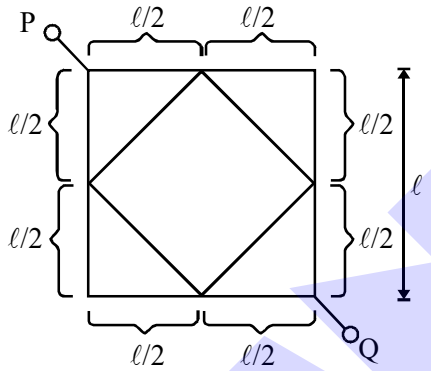
19. In the diagram resistance between any two junctions is R. Equivalent resistance across terminals A and B is :-



(A) $\frac{11R}{7}$ (B) $\frac{18R}{11}$ (C) $\frac{7R}{11}$ (D) $\frac{11R}{18}$

CE0064

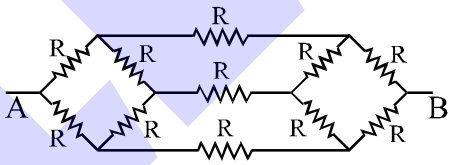
20. If the wire has resistivity ρ and cross sectional area A, the equivalent resistance between P and Q is :-



(A) $\frac{\rho\ell}{\sqrt{2}A}$ (B) $\frac{\sqrt{2}\rho\ell}{A}$ (C) $\frac{2\rho\ell}{A}$ (D) $\frac{\rho\ell}{A}$

CE0072

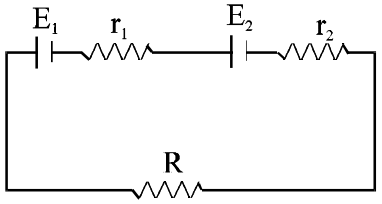
21. The equivalent resistance between the terminal points A and B in the network shown in figure is :-



(A) $\frac{7R}{5}$ (B) $\frac{5R}{6}$ (C) $\frac{7R}{12}$ (D) $\frac{5R}{12}$

CE0073

22. Under what condition current passing through the resistance R can be increased by short circuiting the battery of emf E_2. The internal resistances of the two batteries are r_1 and r_2 respectively.

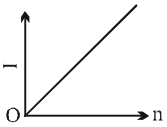
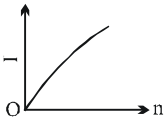
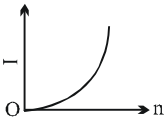
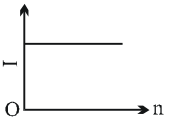
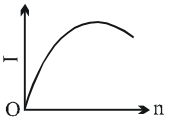


(A) $E_2r_1 > E_1(R + r_2)$ (B) $E_1r_2 > E_2(R + r_1)$
(C) $E_2r_2 > E_1(R + r_2)$ (D) $E_1r_1 > E_2(R + r_1)$

CE0059

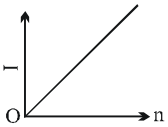
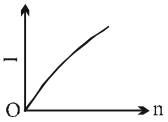
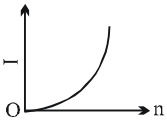
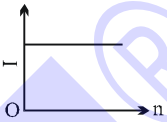
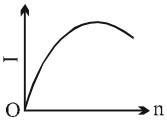
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23. A battery consists of a variable number n of identical cells having internal resistance connected in series. The terminals of the battery are short circuited and the current I measured. Which one of the graph below shows the relationship between I and n?

(A)  (B)  (C)  (D)  (E) 

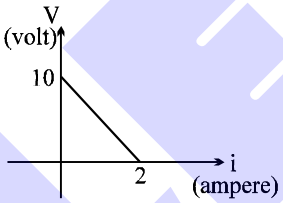
CE0060

24. In **previous problem**, if the cell had been connected in parallel (instead of in series) which of the above graphs would have shown the relationship between total current I and n?

(A)  (B)  (C)  (D)  (E) 

CE0061

25. A battery of emf E and internal resistance r is connected across a resistance R. Resistance R can be adjusted to any value greater than or equal to zero. A graph is plotted between the current (i) passing through the resistance and potential difference (V) across it. Select the correct alternative.



(A) internal resistance of battery is 5Ω
(B) emf of the battery is 20V
(C) maximum current which can be taken from the battery is 4A
(D) V- i graph can never be a straight line as shown in figure.

CE0068

D. Heating effect due to current and Electrical Instruments

26. Power generated across a uniform wire connected across a supply is H. If the wire is cut into n equal parts and all the parts are connected in parallel across the same supply, the total power generated in the wire is

(A) $\frac{H}{n^2}$ (B) n^2H (C) nH (D) $\frac{H}{n}$

CE0076

27. When electric bulbs of same power, but different marked voltage are connected in series across the power line, their brightness will be :
(A) proportional to their marked voltage
(B) inversely proportional to their marked voltage
(C) proportional to the square of their marked voltage
(D) inversely proportional to the square of their marked voltage

CE0077

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grow\-Current Elec.-Copy\Eng\03_Current Electricity\01_Eng.p65

28. Two bulbs rated (25 W – 220V) and (100W – 220V) are connected in series to a 440 V line. Which one is likely to fuse?

(A) 25 W bulb (B) 100 W bulb (C) both bulbs (D) none

CE0078

29. Rate of dissipation of Joule's heat in resistance per unit volume is (symbols have usual meaning)

(A) σE (B) σJ (C) J E (D) None

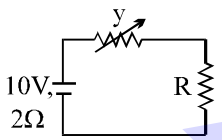
CE0079

30. Two bulbs one of 200 volts, 60 watts & the other of 200 volts, 100 watts are connected in series to a 200 volt supply. The power consumed will be

(A) 37.5 watt (B) 160 watt (C) 62.5 watt (D) 110 watt

CE0081

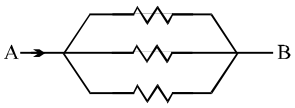
31. In the figure shown the power generated in y is maximum when y = 5Ω. Then R is :-



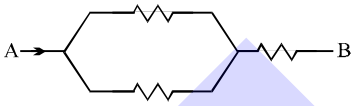
(A) 2 Ω (B) 6 Ω (C) 5 Ω (D) 3 Ω

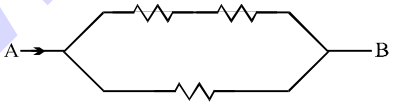
CE0082

32. Arrange the order of power dissipated in the given circuits, if the same current is passing through all circuits and each resistor is 'r' **[IIT-JEE' 2003 (Scr)]**

(1) 

(2) 

(3) 

(4) 

(A) $P_2 > P_3 > P_4 > P_1$ (B) $P_3 > P_2 > P_4 > P_1$ (C) $P_4 > P_3 > P_2 > P_1$ (D) $P_1 > P_2 > P_3 > P_4$

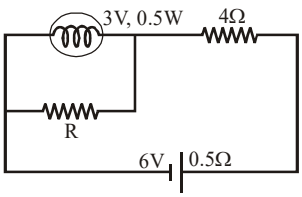
CE0083

33. A rigid container with thermally insulated walls contains a coil of resistance 100 Ω, carrying current 1 A. Change in internal energy after 5 min will be **[IIT-JEE 2005]**

(A) zero (B) 10 kJ (C) 20 kJ (D) 30 kJ

CE0084

34. The value of the resistance R in the circuit shown below so that electric bulb consumes the rated power is :-



(A) 4 ohm (B) 6 ohm (C) 8 ohm (D) 10 ohm

CE0085

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Capa\Eng\03_Current Electricity\01_Eng.p65

35. A variable load R is connected to a voltage source of internal resistance r. Then choose the **INCORRECT** statement out of the following :-

(A) If R = r, maximum power is transferred to the load

(B) If current is maximum, power transfer to load is also maximum

(C) If R << r, the voltage source supplied a fixed current to the load

(D) Power supplied to load is minimum if load is either too low or too high

CE0086

36. The variation of current (I) and voltage (V) is as shown in figure A. The variation of power P with current *I* is best shown by which of the following graph

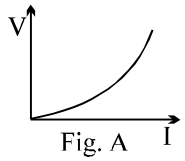
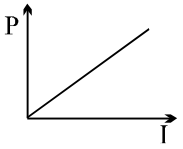
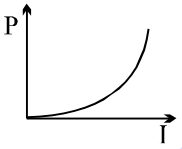
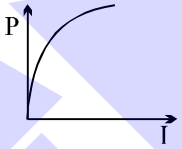
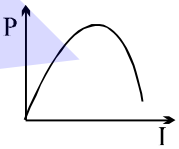


Fig. A

(A) 

(B) 

(C) 

(D) 

CE0087

37. Incandescent bulbs are designed by keeping in mind that the resistance of their filament increases with the increase in temperature. If at room temperature, 100 W, 60 W and 40 W bulbs have filament resistance R_{100}, R_{60} and R_{40}, respectively, the relation between these resistances is **[IIT-JEE 2010]**

(A) $\frac{1}{R_{100}} = \frac{1}{R_{40}} + \frac{1}{R_{60}}$

(B) $R_{100} = R_{40} + R_{60}$

(C) $R_{100} > R_{60} > R_{40}$

(D) $\frac{1}{R_{100}} > \frac{1}{R_{60}} > \frac{1}{R_{40}}$

CE0152

38. A galvanometer has a resistance of 20Ω and reads full-scale when 0.2 V is applied across it. To convert it into a 10 A ammeter, the galvanometer coil should have a

(A) 0.01Ω resistor connected across it

(B) 0.02Ω resistor connected across it

(C) 200Ω resistor connected in series with it

(D) 2000 Ω resistor connected in series with it

CE0088

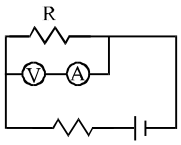
39. A galvanometer coil has a resistance 90 Ω and full scale deflection current 10 mA. A 910Ω resistance is connected in series with the galvanometer to make a voltmeter. If the least count of the voltmeter is 0.1V, the number of divisions on its scale is

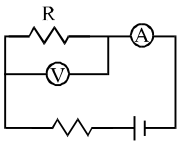
(A) 90 (B) 91 (C) 100 (D) none

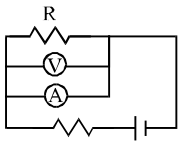
CE0089

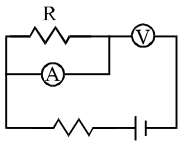
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grow\-Current Elec.-Copy\Eng\03_Current Electricity\01_Eng.p65

40. Which of the following wiring diagrams could be used to experimentally determine R using ohm's law? Assume an ideal voltmeter and an ideal ammeter.

(A) 

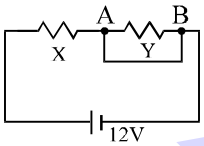
(B) 

(C) 

(D) 

CE0090

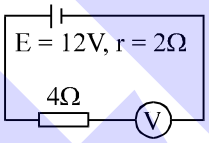
41. When an ammeter of negligible internal resistance is inserted in series with circuit it reads 1A. When the voltmeter of very large resistance is connected across X it reads 1V. When the point A and B are shorted by a conducting wire, the voltmeter measures 10 V across the battery. The internal resistance of the battery is equal to :-



(A) zero (B) 0.5 Ω (C) 0.2 Ω (D) 0.1 Ω

CE0091

42. By error, a student places moving-coil voltmeter V (nearly ideal) in series with the resistance in a circuit in order to read the current, as shown. The voltmeter reading will be



(A) 0 (B) 4V (C) 6V (D) 12V

CE0092

43. In a balanced wheat stone bridge, current in the galvanometer is zero. It remains zero when:
[1] battery emf is increased
[2] all resistances are increased by 10 ohms
[3] all resistances are made five times
[4] the battery and the galvanometer are interchanged
(A) only [1] is correct (B) [1], [2] and [3] are correct
(C) [1], [3] and [4] are correct (D) [1] and [3] are correct

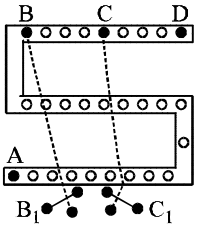
CE0094

44. A resistance R is to be measured using a meter bridge. Student chooses the standard resistance S to be 100Ω. He finds the null point at $l_1 = 2.9$ cm. He is told to attempt to improve the accuracy. Which of the following is a useful way?
(A) He should measure l_1 more accurately.
(B) He should change S to 1000Ω and repeat the experiment.
(C) He should change S to 3Ω and repeat the experiment.
(D) He should give up hope of a more accurate measurement with a meter bridge.

CE0097

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Copo\Eng\03_Current Electricity\01_Eng.p65

45. For the post office box arrangement to determine the value of unknown resistance, the unknown resistance should be connected between **[IIT-JEE' 2004 (Scr)]**



(A) *B* and *C* (B) *C* and *D* (C) *A* and *D* (D) B_1 and C_1

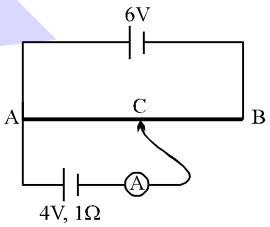
CE0099

46. A potentiometer wire has length 10 m and resistance 10Ω. It is connected to a battery of EMF 11 volt and internal resistance 1 Ω, then the potential gradient in the wire is :-

(A) 10 V/m (B) 1 V/m (C) 0.1 V/m (D) none

CE0100

47. A 6 V battery of negligible internal resistance is connected across a uniform wire of length 1 m. The positive terminal of another battery of emf 4V and internal resistance 1 Ω is joined to the point A as shown in figure. The ammeter shows zero deflection when the jockey touches the wire at the point C. The AC is equal to :-

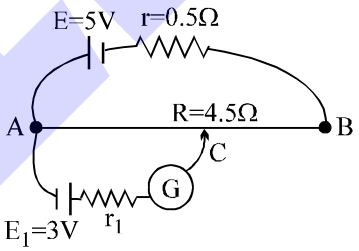


(A) 2/3 m (B) 1/3 m

(C) 3/5 m (D) 1/2 m

CE0101

48. In the given potentiometer circuit length of the wire AB is 3 m and resistance is R = 4.5 Ω. The length AC for no deflection in galvanometer is :-



(A) 2 m (B) 1.8 m (C) dependent on r_1 (D) none of these

CE0102

49. Two cells of emf's approximately 5V and 10V are to be accurately compared using a potentiometer of length 400cm.

(A) The battery that runs the potentiometer should have voltage of 8V.

(B) The battery of potentiometer can have a voltage of 15V and *R* adjusted so that the potential drop across the wire slightly exceeds 10V.

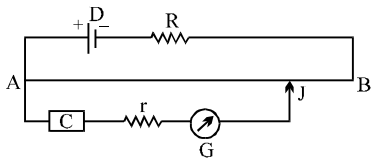
(C) The first portion of 50 cm of wire itself should have a potential drop of 10V.

(D) Potentiometer is usually used for comparing resistances and not voltages.

CE0104

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grow\Current Elec.-Copy\Eng\03_Current Electricity\01_Eng.p65

50. In the given potentiometer circuit, the resistance of the potentiometer wire AB is R_0. C is a cell of internal resistance r. The galvanometer G does not give zero deflection for any position of the jockey J. Which of the following cannot be a reason for this ?



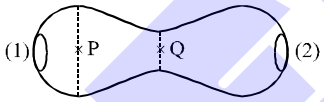
(A) $r > R_0$
(B) $R >> R_0$
(C) emf of C > emf of D
(D) The negative terminal of C is connected to A.

CE0105

MULTIPLE CORRECT TYPE QUESTIONS

A. Microscopic Ohm's Law

51. A metallic conductor of irregular cross-section is as shown in the figure. A constant potential difference is applied across the ends (1) and (2). Then :



(A) the current at the cross-section P equals the current at the cross-section Q
(B) the electric field intensity at P is less than that at Q.
(C) the rate of heat generated per unit time at Q is greater than that at P
(D) the number of electrons crossing per unit area of cross-section at P is less than that at Q.

CE0108

52. A current passes through an ohmic conductor of nonuniform cross section. Which of the following quantities are independent of the cross-section?
(A) the charge crossing in a given time interval.
(B) drift speed
(C) current density
(D) free-electron density

CE0109

B. Kirchoff's Laws

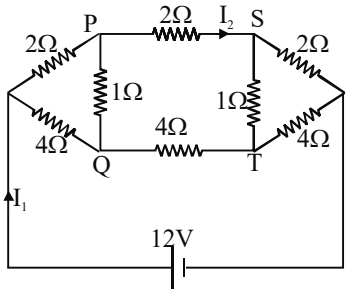
53. A battery is of emf E is being charged from a charger such that positive terminal of the battery is connected to terminal A of charger and negative terminal of the battery is connected to terminal B of charger. The internal resistance of the battery is r.
(A) Potential difference across points A and B must be more than E.
(B) A must be at higher potential than B
(C) In battery, current flows from positive terminal to the negative terminal
(D) No current flows through battery

CE0110

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Copy\Eng\03_Current Electricity\01_Eng.p65

C. Equivalent Resistance and equivalent of battery

54. For the resistance network shown in the figure, choose the correct option(s). **[IIT-JEE 2012]**



(A) the current through PQ is zero (B) $I_1 = 3A$

(C) The potential at S is less than that at Q (D) $I_2 = 2A$

CE0158

D. Heating effect due to current and Electrical Instruments

55. Two identical fuses are rated at 10A. If they are joined

(A) in parallel, the combination acts as a fuse of rating 20A

(B) in parallel, the combination acts as a fuse of rating 5A

(C) in series, the combination acts as a fuse of rating 10A.

(D) in series, the combination acts as a fuse of rating 20A.

CE0111

56. A micrometer has a resistance of 100Ω and a full scale range of 50μA. It can be used as a voltmeter or a higher range ammeter provided a resistance is added to it. Pick the correct range and resistance combination(s).

(A) 50 V range with 10 kΩ resistance in series.

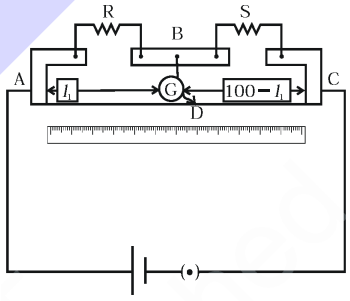
(B) 10 V range with 200 kΩ resistance in series.

(C) 5 mA range with 1 Ω resistance in parallel.

(D) 10 mA range with 1 kΩ resistance in parallel.

CE0112

57. In a meter bridge the point D is a neutral point as shown in figure.



(A) The meter bridge can have no other neutral point for this set of resistances.

(B) When the jockey contacts a point on meter wire left of D, current flows to B from the wire through galvanometer.

(C) When the jockey contacts a point on the meter wire to the right of D, current flows from B to the wire through galvanometer.

(D) When *R* is increased, the neutral point shifts to left.

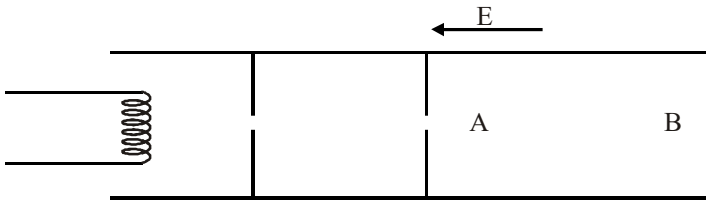
CE0114

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\Current Elec.-Copa\Eng\03_Current Electricity\01_Eng.p65

MATRIX MATCH TYPE QUESTION

A. Microscopic Ohm's Law

58. Electrons are emitted by a hot filament and are accelerated by an electric field as shown in figure. The two stops at the left ensure that the electron beam has a uniform cross-section. Match the entries of column-I with column-II as electron move from A to B :

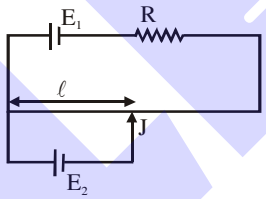


	Column-I		**Column-II**
(A)	Speed of an electron	(P)	Inreases
(B)	Number of free electrons per unit volume	(Q)	Decreases
(C)	Current density	(R)	Remains same
(D)	Electric potential	(S)	any of the above is possible

CE0115

D. Heating effect due to current and Electrical Instruments

59. In the potentiometer arrangement shown in figure, null point is obtained at length ℓ.



	Column-I		**Column-II**
(A)	If E_1 is increased	(P)	ℓ should increase
(B)	If R is increased	(Q)	ℓ should decrease
(C)	If E_2 is increased	(R)	ℓ should remain the same to again get the null point

CE0116

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Capa\Eng\03_Current Electricity\01_Eng.p65

EXERCISE # S

A. Microscopic Ohm's Law

1. A copper wire of length L, and cross section area A carries a current I. If the specific resistance of copper is ρ, the electric field in the wire is _ _ _ _ _ _ .

CE0001

2. A copper wire carries a current density j (= current per unit area). Assuming that n = No. of free electrons per unit volume, e = electronic charge, <v> = average speed due to thermal agitation. The distance which will be covered by an electron during its displacement l along the wire _ __ _ _.

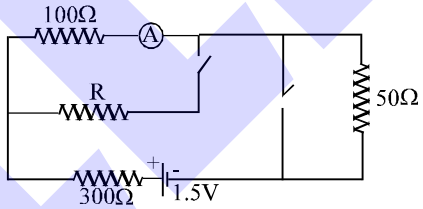
CE0002

3. The total momentum of electrons in a straight wire of length ℓ carrying a current I is _ _ _ _ _ _ (mass of electron = m_e, charge of electron = e)

CE0003

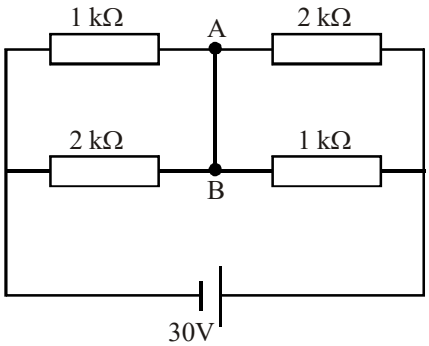
B. Kirchoff's Laws

4. In the circuit shown in figure the reading of ammeter is the same with both switches open as with both closed. Then find the resistance R. (ammeter is ideal)



CE0006

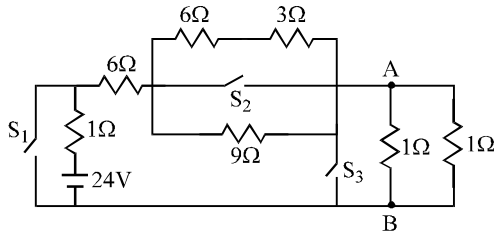
5. Find the current (in mA) in the wire between points A and B.



CE0007

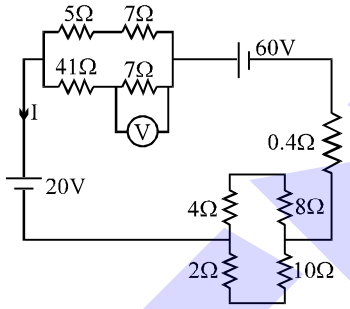
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grow\Current Elec.-Copy\Eng\03_Current Electricity\01_Eng.p65

6. If the switches S_1, S_2 and S_3 in the figure are arranged such that current through the battery is minimum, find the voltage across points A and B.



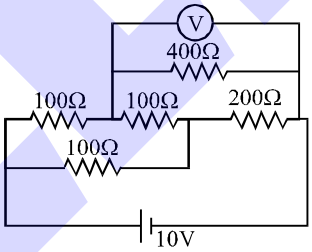
CE0008

7. Find the current I & voltage V in the circuit shown.



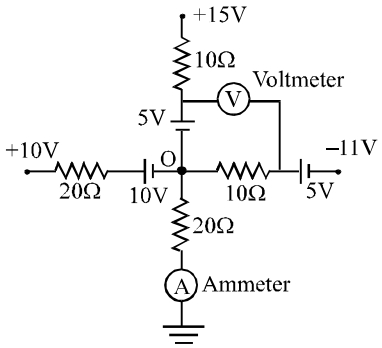
CE0009

8. An electrical circuit is shown in the figure. Calculate the potential difference across the resistance of 400 ohm, as will be measured by the voltmeter V of resistance 400 ohm, either by applying Kirchhoff's rules or otherwise.



CE0010

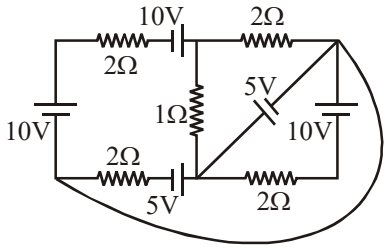
9. The potential of certain points in the circuit are maintained at the values indicated. The Voltmeter and Ammeter are ideal. Find the potential of the cross junction point in the circuit (at center O) and the readings of Voltmeter and Ammeter. All cells are ideal.



CE0014

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Capa\Eng\03_CurrentElectricity\01_Eng.p65

10. In the given circuit diagram, the current through the 1Ω resistor is given by I amp. Fill 2 I in OMR sheet.



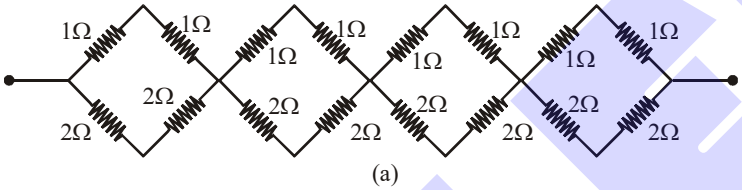
CE0015

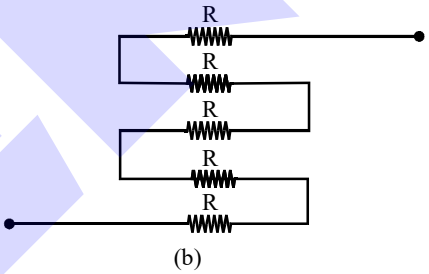
C. Equivalent Resistance and equivalent of battery

11. (a) Given n resistors each of resistance R, how will you combine them to get the (i) maximum (ii) minimum effective resistance? What is the ratio of the maximum to minimum resistance?

(b) Given the resistances of 1Ω, 2Ω, 3Ω, how will be combine them to get an equivalent resistance of (i) (11/3) Ω (ii) (11/5) Ω (iii) 6Ω (iv) (6/11) Ω ?

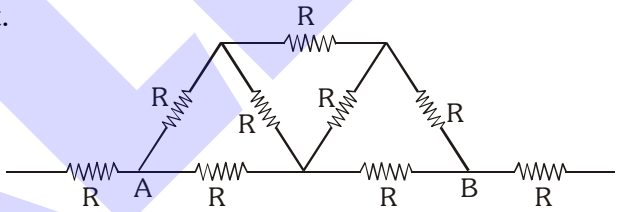
(c) Determine the equivalent resistance of networks shown in figures





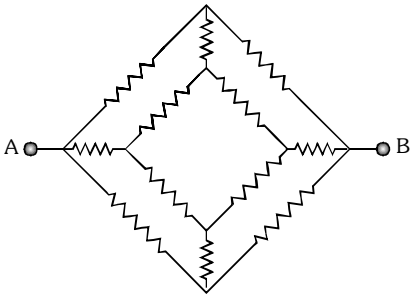
CE0005

12. Find the effective resistance of the network (see figure) between the points A and B. Where R is the resistance of each part.



CE0319

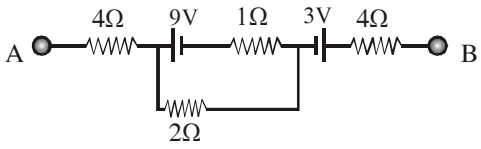
13. The figure shows a network of resistor each heaving value 12Ω. Find the equivalent resistance between points A and B.



CE0320

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14. In the circuit shown in figure potential difference between point A and B is 16V. Find the current passing through 2Ω resistance.

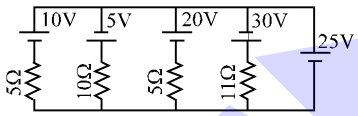


CE0011

15. An enquiring physics student connects a cell to a circuit and measures the current drawn from the cell to I_1. When he joins a second identical cell in series with the first, the current becomes I_2. When the cells are connected are in parallel, the current through the circuit is I_3. Show that relation between the current is $3\, I_3 I_2 = 2\, I_1 (I_2 + I_3)$

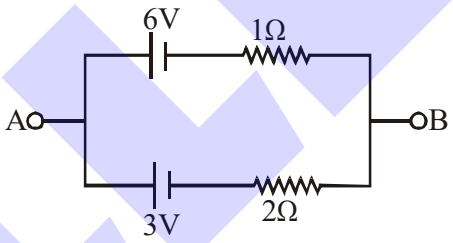
CE0012

16. Find the current through 25V cell & power supplied by 20V cell in the figure shown.



CE0019

17. Two batteries of different emfs and different internal resistances are connected as shown. The voltage across AB in volts is **[IIT-JEE 2011]**



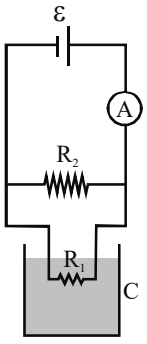
CE0156

D. Heating effect due to current and Electrical Instruments

18. If a cell of constant E.M.F. produces the same amount of the heat during the same time in two independent resistors R_1 and R_2, when they are separately connected across the terminals of the cell, one after the another, find the internal resistance of the cell.

CE0016

19. The coil of a calorimeter C has a resistance of $R_1 = 60\Omega$. The coil R_1 is connected to the circuit as shown in figure. What is the rise in temperature (°C) of 240 grams of water poured into the calorimeter when it is heated for 7 minutes during which a current flows through the coil and the ammeter shows 3A? The resistance $R_2 = 30\Omega$. [Disregard the resistances of the battery and the ammeter, and the heat losses and heat capacity of the calorimeter and the resistor and specific heat of water = 4200 J/kg°C]



CE0017

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Capa\Eng\03_Current Electricity\01_Eng.p65

20. An electric kettle has two windings. When one of them is switched on, the water in the kettle begins to boil in 15 minutes, and when the other is switched on it takes 30 minutes for water to boil. If the two windings are joined in series and switched on, water in the kettle begin to boil in $\frac{\alpha}{4}$ hr. Assuming no heat loss to the surrounding fill the value of α in OMR sheet.

CE0018

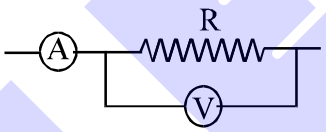
21. A person decides to use his bath tub water to generate electric power to run a 40 watt bulb. The bath tub is located at a height of 10 m from the ground & it holds 200 litres of water. If we install a water driven wheel generator on the ground, at what rate should the water drain from the bath tub to light bulb? How long can we keep the bulb on, if the bath tub was full initially. The efficiency of generator is 90%.($g = 10 m/s^2$)

CE0020

22. When two identical batteries of internal resistance 1Ω each are connected in series across a resistor R, the rate of heat produced in R is J_1. When the same batteries are connected in parallel across R, the rate is J_2. If $J_1 = 2.25\ J_2$ then the value of R in Ω is **[IIT-JEE 2010]**

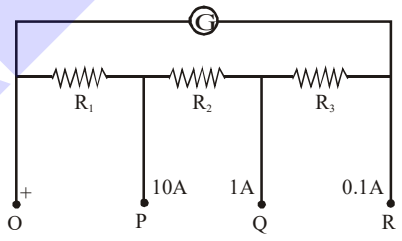
CE0155

23. A part of a circuit is shown in figure. Here reading of ammeter is 5 ampere and voltmeter is 96V & voltmeter resistance is 480 ohm. Then find the resistance R



CE0021

24. The resistance of the galvanometer G in the circuit is 25Ω. The meter deflects full scale for a current of 10 mA. The meter behaves as an ammeter of three different ranges. The range is 0–10A, if the terminals O and P are taken; range is 0–1 A between O and Q; range is 0–0.1 A between O and R. Calculate the resistance R_1, R_2 and R_3.



CE0022

25. A galvanometer having 50 divisions provided with a variable shunt is used to measure the current as an ammeter when connected in series with a resistance of 90 Ω and a battery of internal resistance 10Ω. It is observed that when the shunt resistance are 10Ω, 50Ω, respectively the deflection are respectively 9 & 30 divisions. What is the resistance of the galvanometer? Further if the full scale deflection of the galvanometer movement is 300 mA, find the emf of the cell.

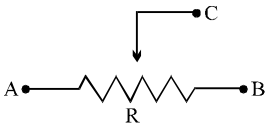
CE0023

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26. Draw the circuit for experimental verification of Ohm's law using a source of variable D.C. voltage, a main resistance of 100 Ω, two galvanometers and two resistances of values 10^6 Ω and 10^{-3} Ω respectively. Clearly show the positions of the voltmeter and the ammeter. **[IIT-JEE' 2004]**

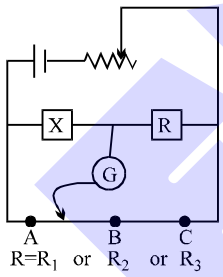
CE0024

27. How a battery is to be connected so that shown rheostat will behave like a potential divider? Also indicate the points about which output can be taken. **[IIT-JEE' 2003]**



CE0025

28. An unknown resistance X is to be determined using resistances R_1, R_2 or R_3. Their corresponding null points are A, B and C. Find which of the above will give the most accurate reading and why? **[IIT-JEE 2005]**



CE0026

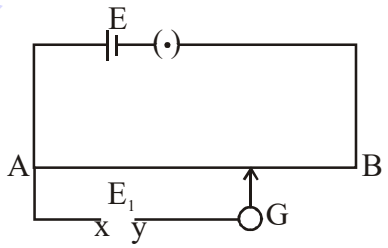
29. While doing an experiment with potentiometer it was found that the deflection is one sided and two casses are possible

(i) the deflection decreased while moving from one end A of the wire to the end B;

(ii) the deflection increased. while the jockey was moved towards the end B. Then

(a) Which terminal +or –ve of the cell E_1, is connected at X in case (i) and how is E_1 related to E?

(b) Which terminal of the cell E_1 is connected at X in case (ii)?



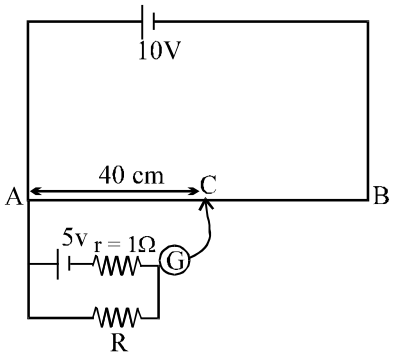
CE0028

30. In a potentiometer arrangement, a cell of emf 1.25 V gives a balance point at 35.0 cm length of the wire. If the cell is replaced by another cell and the balance point shifts to 63.0 cm, what is the emf of the second cell ?

CE0029

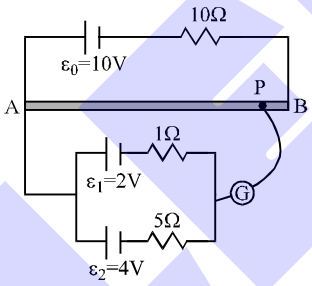
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Copo\Eng\03_CurrentElectricity\01_Eng.p65

31. A potentiometer wire AB is 100 cm long and has a total resistance of 10 ohm. If the galvanometer shows zero deflection at the position C, then find the value of unknown resistance R.



CE0030

32. A battery of emf ε_0 = 10 V is connected across a 1 m long uniform wire having resistance 10Ω/m. Two cells of emf ε_1 = 2V and ε_2 = 4V having internal resistances 1Ω and 5Ω respectively are connected as shown in the figure. If a galvanometer shows no deflection at the point P, find the distance of point P from the point A.



CE0031

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grow\Current Elec.-Copy\Eng\03_Current Electricity\01_Eng.p65

EXERCISE-JM

1. The supply voltage to a room is 120V. The resistance of the lead wires is 6Ω. A 60 W bulb is already switched on. What is the decrease of voltage across the bulb, when a 240 W heater is switched on in parallel to the bulb? **[JEE-Main2013]**

(1) zero Volt (2) 2.9 Volt (3) 13.3 Volt (4) 10.04 Volt

CE0136

2. This question has Statement I and Statement II. Of the four choice given after the Statements, choose the one that best describes the two Statements. **[JEE-Main 2013]**

Statement-I : Higher the range, greater is the resistance of ammeter.

Statement-II : To increase the range of ammeter, additional shunt needs to be used across it.

(1) Statement-I is true, Statement-II is true, Statement-II is the **correct** explanation of Statement-I

(2) Statement-I is true, Statement-II is true, Statement-II is **not** the correct explanation of Statement-I.

(3) Statement-I is **true,** Statement-II is false.

(4) Statement-I is **false,** Statement-II is true.

CE0137

3. In a large building, there are 15 bulbs of 40 W, 5 bulbs of 100 W, 5 fans of 80 W and 1 heater of 1kW. The voltage of the electric mains is 220 V. The minimum capacity of the main fuse of the building will be : **[JEE-Main 2014]**

(1) 12 A (2) 14 A (3) 8 A (4) 10 A

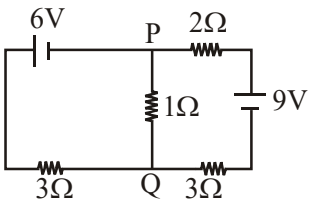
CE0138

4. When 5V potential difference is applied across a wire of length 0.1 m, the drift speed of electrons is 2.5×10^{-4} ms^{-1}. If the electron density in the wire is 8×10^{28} m^{-3}, the resistivity of the material is close to :- **[JEE-Main 2015]**

(1) 1.6×10^{-6} Ωm (2) 1.6×10^{-5} Ωm (3) 1.6×10^{-8} Ωm (4) 1.6×10^{-7} Ωm

CE0139

5. In the circuit shown, the current in the 1Ω resistor is :- **[JEE-Main 2015]**



(1) 0.13 A, from Q to P (2) 0.13 A, from P to Q

(3) 1.3 A, from P to Q (4) 0A

CE0140

6. A galvanometer having a coil resistance of 100Ω gives a full scale deflection, when a current of 1 mA is passed through it. The value of the resistance, which can convert this galvanometer into ammeter giving a full scale deflection for a current of 10A, is :- **[JEE-Main 2016]**

(1) 3Ω (2) 0.01Ω (3) 2Ω (4) 0.1Ω

CE0141

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Capa\Eng\03_Current Electricity\01_Eng.p65

7. A 50Ω resistance is connected to a battery of 5V. A galvanometer of resistance 100 Ω is to be used as an ammeter to measure current through the resistance, for this a resistance r_s is connected to the galvanometer. Which of the following connections should be employed if the measured current is within 1% of the current without the ammeter in the circuit ? **[JEE-Mains (Online) 2016]**

(1) $r_s = 1\Omega$ in series with galvanometer

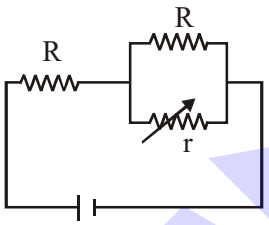
(2) $r_s = 0.5\ \Omega$ in parallel with the galvanometer

(3) $r_s = 0.5\ \Omega$ in series with the galvanometer

(4) $r_s = 1\ \Omega$ in parallel with galvanometer

CE0142

8. In the circuit shown, the resistance r is a variable resistance. If for r = f R, the heat generation in r is maximum then the value of f is : **[JEE-Mains (Online) 2016]**



(1) 1 (2) $\frac{1}{4}$ (3) $\frac{1}{2}$ (4) $\frac{3}{4}$

CE0143

9. Which of the following statements is false ? **[JEE-Main 2017]**

(1) A rheostat can be used as a potential divider

(2) Kirchhoff's second law represents energy conservation

(3) Wheatstone bridge is the most sensitive when all the four resistances are of the same order of magnitude.

(4) In a balanced wheatstone bridge if the cell and the galvanometer are exchanged, the null point is disturbed.

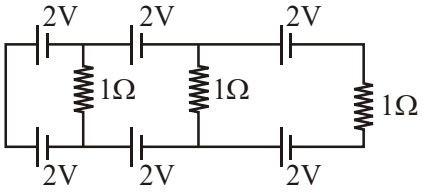
CE0146

10. When a current of 5 mA is passed through a galvanometer having a coil of resistance 15 Ω, it shows full scale deflection. The value of the resistance to be put in series with the galvanometer to convert it into to voltmeter of range 0 – 10 V is :- **[JEE-Main 2017]**

(1) $2.535 \times 10^3\ \Omega$ (2) $4.005 \times 10^3\ \Omega$ (3) $1.985 \times 10^3\ \Omega$ (4) $2.045 \times 10^3\ \Omega$

CE0147

11. In the above circuit the current in each resistance is :- **[JEE-Main 2017]**



(1) 0.5 A (2) 0 A (3) 1 A (4) 0.25 A

CE0148

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grow\3-Current Elec.-Copa\Eng\03_Current Electricity\01_Eng.pt65

12. In a potentiometer experiment, it is found that no current passes through the galvanometer when the terminals of the cell are connected across 52 cm of the potentiometer wire. If the cell is shunted by a resistance of 5 Ω, a balance is found when the cell is connected across 40 cm of the wire. Find the internal resistance of the cell. **[JEE-Main 2018]**

(1) 1.5 Ω (2) 2 Ω (3) 2.5 Ω (4) 1 Ω

CE0149

13. On interchanging the resistances, the balance point of a meter bridge shifts to the left by 10 cm. The resistance of their series combination is 1 kΩ. How much was the resistance on the left slot before interchanging the resistances ? **[JEE-Main 2018]**

(1) 505 kΩ (2) 550 kΩ (3) 910 kΩ (4) 990 kΩ

CE0150

14. Two batteries with e.m.f 12 V and 13 V are connected in parallel across a load resistor of 10 Ω. The internal resistances of the two batteries are 1 Ω and 2 Ω respectively. The voltage across the load lies between. **[JEE-Main 2018]**

(1) 11.5 V and 11.6 V (2) 11.4 V and 11.5 V

(3) 11.7 V and 11.8 V (4) 11.6 V and 11.7 V

CE0151

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

15. A carbon resistance has a following colour code. What is the value of the resistance ? **[JEE-Main-2019_Jan]**

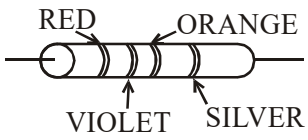


(1) 1.64 MΩ ± 5% (2) 530 kΩ ± 5%

(3) 64 kΩ ± 10% (4) 5.3 MΩ ± 5%

CE0196

16. A resistance is shown in the figure. Its value and tolerance are given respectively by: **[JEE-Main-2019_Jan]**



(1) 27 KΩ, 20% (2) 270 KΩ, 5%

(3) 270 KΩ, 10% (4) 27 KΩ, 10%

CE0197

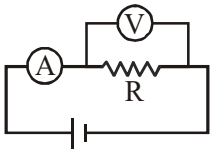
17. A copper wire is stretched to make it 0.5% longer. The percentage change in its electrical resistance if its volume remains unchanged is: **[JEE-Main-2019_Jan]**

(1) 2.5% (2) 0.5% (3) 1.0% (4) 2.0%

CE0198

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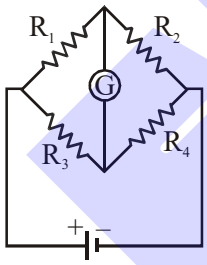
18. The actual value of resistance R, shown in the figure is 30Ω. This is measured in an experiment as shown using the standard formula $R = \frac{V}{I}$, where V and I are the readings of the voltmeter and ammeter, respectively. If the measured value of R is 5% less, then the internal resistance of the voltmeter is : **[JEE-Main-2019_Jan]**



(1) 350Ω (2) 570Ω (3) 35 Ω (4) 600 Ω

CE0199

19. The Wheatstone bridge shown in Fig. here, gets balanced when the carbon resistor used as R_1 has the colour code (Orange, Red, Brown). The resistors R_2 and R_4 are 80Ω and 40Ω, respectively. Assuming that the colour code for the carbon resistors gives their accurate values, the colour code for the carbon resistor, used as R_3, would be : **[JEE-Main-2019_Jan]**



(1) Red, Green, Brown (2) Brown, Blue, Brown
(3) Grey, Black, Brown (4) Brown, Blue, Black

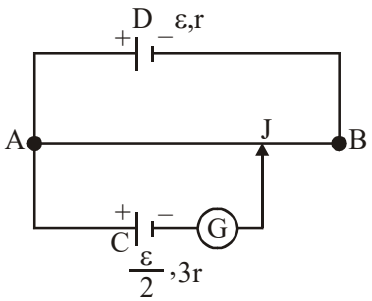
CE0200

20. A 2 W carbon resistor is color coded with green, black, red and brown respectively. The maximum current which can be passed through this resistor is : **[JEE-Main-2019_Jan]**

(1) 63 mA (2) 0.4 mA (3) 100 mA (4) 20 mA

CE0201

21. A potentiometer wire AB having length L and resistance 12 r is joined to a cell D of emf ε and internal resistance r. A cell C having emf ε/2 and internal resistance 3r is connected. The length AJ at which the galvanometer as shown in fig. shows no deflection is : **[JEE-Main-2019_Jan]**



(1) $\frac{5}{12}$L (2) $\frac{11}{24}$L (3) $\frac{11}{12}$L (4) $\frac{13}{24}$L

CE0202

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Copy\Eng\03_Current Electricity\01_Eng.p65

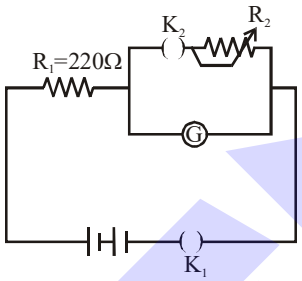
22. Two equal resistance when connected in series to a battery, consume electric power of 60 W. If these resistances are now connected in parallel combination to the same battery, the electric power consumed will be : **[JEE-Main-2019_Jan]**

(1) 60 W (2) 240 W

(3) 30 W (4) 120 W

CE0203

23. The galvanometer deflection, when key K_1 is closed but K_2 is open, equals θ_0 (see figure). On closing K_2 also and adjusting R_2 to 5Ω, the deflection in galvanometer becomes $\frac{\theta_0}{5}$. The resistance of the galvanometer is, then, given by [Neglect the internal resistance of battery]: **[JEE-Main-2019_Jan]**

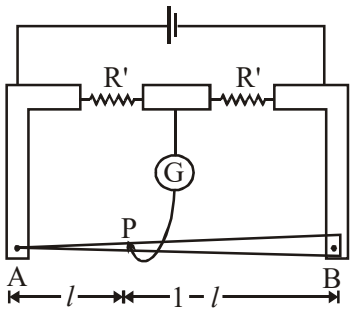


(1) 12Ω (2) 25Ω

(3) 5Ω (4) 22Ω

CE0204

24. In a meter bridge, the wire of length 1 m has a non-uniform cross-section such that, the variation $\frac{dR}{d\ell}$ of its resistance R with length ℓ is $\frac{dR}{d\ell} \propto \frac{1}{\sqrt{\ell}}$. Two equal resistances are connected as shown in the figure. The galvanometer has zero deflection when the jockey is at point P. What is the length AP? **[JEE-Main-2019_Jan]**

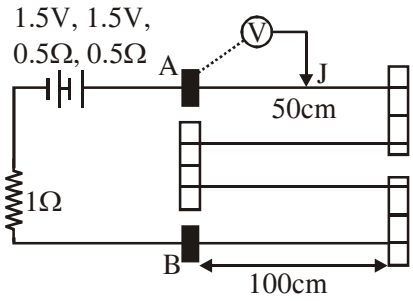


(1) 0.25 m (2) 0.3m (3) 0.35 m (4) 0.2 m

CE0205

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Capa\Eng\03_Current Electricity\01_Eng.p65

25. In the circuit shown, a four-wire potentiometer is made of a 400 cm long wire, which extends between A and B. The resistance per unit length of the potentiometer wire is r = 0.01 Ω/cm. If an ideal voltmeter is connected as shown with jockey J at 50 cm from end A, the expected reading of the voltmeter will be :- **[JEE-Main-2019_April]**



(1) 0.20 V (2) 0.25 V (3) 0.75 V (4) 0.50V

CE0206

26. In a conductor, if the number of conduction electrons per unit volume is 8.5×10^{28} m^{-3} and mean free time is 25*f*s (femto second), it's approximate resistivity is :- ($m_e = 9.1 \times 10^{-31}$ kg) **[JEE-Main-2019_April]**

(1) 10^{-5} Ωm (2) 10^{-6} Ωm (3) 10^{-7} Ωm (4) 10^{-8} Ωm

CE0207

27. Space between two concentric conducting spheres of radii a and b (b > a) is filled with a medium of resistivity ρ. The resistance between the two spheres will be : **[JEE-Main-2019_April]**

(1) $\frac{\rho}{4\pi}\left(\frac{1}{a}-\frac{1}{b}\right)$ (2) $\frac{\rho}{2\pi}\left(\frac{1}{a}-\frac{1}{b}\right)$ (3) $\frac{\rho}{2\pi}\left(\frac{1}{a}+\frac{1}{b}\right)$ (4) $\frac{\rho}{4\pi}\left(\frac{1}{a}+\frac{1}{b}\right)$

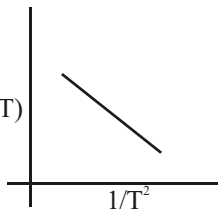
CE0208

28. A current of 5 A passes through a copper conductor (resistivity = 1.7×10^{-8} Ωm) of radius of cross-section 5 mm. Find the mobility of the charges if their drift velocity is 1.1×10^{-3} m/s. **[JEE-Main-2019_April]**

(1) 1.3 m^2/Vs (2) 1.5 m^2/Vs (3) 1.8 m^2/Vs (4) 1.0 m^2/Vs

CE0209

29. In an experiment, the resistance of a material is plotted as a function of temperature (in some range). As shown in the figure, it is a straight line. One may conclude that : **[JEE-Main-2019_April]**



(1) $R(T) = \frac{R_0}{T^2}$ (2) $R(T) = R_0 e^{-T^2/T_0^2}$ (3) $R(T) = R_0 e^{-T_0^2/T^2}$ (4) $R(T) = R_0 e^{T^2/T_0^2}$

CE0210

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grwi-Current Elec.-Copa\Eng\03_Current Elasticity\01_Eng.p65

30. A moving coil galvanometer, having a resistance G, produces full scale deflection when a current I_g flows through it. This galvanometer can be converted into (i) an ammeter of range 0 to I_0 ($I_0 > I_g$) by connecting a shunt resistance R_A to it and (ii) into a voltmeter of range 0 to V($V = GI_0$) by connecting a series resistance R_V to it. Then, **[JEE-Main-2019_April]**

(1) $R_A R_V = G^2\left(\frac{I_g}{I_0 - I_g}\right)$ and $\frac{R_A}{R_V} = \left(\frac{I_0 - I_g}{I_g}\right)^2$

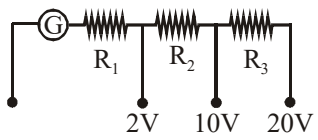
(2) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g}\right)^2$

(3) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \frac{I_g}{(I_0 - I_g)}$

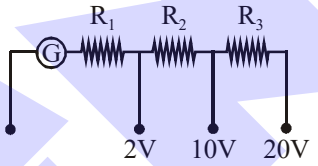
(4) $R_A R_V = G^2\left(\frac{I_0 - I_g}{I_g}\right)$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g}\right)^2$

CE0211

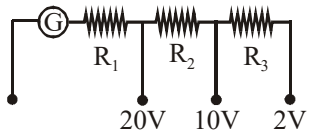
31. A galvanometer of resistance 100Ω has 50 divisions on its scale and has sensitivity of 20 μA/division. It is to be converted to a voltmeter with three ranges, of 0–2 V, 0–10 V and 0–20 V. The appropriate circuit to do so is : **[JEE-Main-2019_April]**

(1)


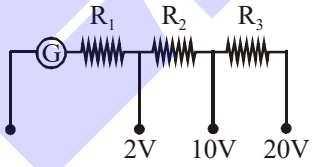
$R_1 = 1900\ \Omega$
$R_2 = 9900\ \Omega$
$R_3 = 19900\ \Omega$

(2)


$R_1 = 2000\ \Omega$
$R_2 = 8000\ \Omega$
$R_3 = 10000\ \Omega$

(3)


$R_1 = 19900\ \Omega$
$R_2 = 9900\ \Omega$
$R_3 = 1900\ \Omega$

(4)


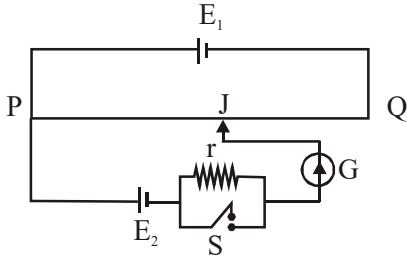
$R_1 = 1900\ \Omega$
$R_2 = 8000\ \Omega$
$R_3 = 10000\ \Omega$

CE0212

32. The series combination of two batteries, both of the same emf 10 V, but different internal resistance of 20Ω and 5Ω, is connected to the parallel combination of two resistors 30 Ω and R Ω. The voltage difference across the battery of internal resistance 20Ω is zero, the value of R (in Ω) is : _____ **[JEE-Main-2020_Jan]**

CE0213

33. A potentiometer wire PQ of 1 m length is connected to a standard cell E_1. Another cell E_2 of emf 1.02 V is connected with a resistance 'r' and switch S (as shown in figure). With switch S open, the null position is obtained at a distance of 49 cm from Q. The potential gradient in the potentiometer wire is: **[JEE-Main-2020_Sep]**



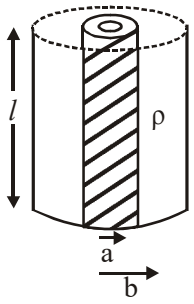
(1) 0.02 V/cm (2) 0.04 V/cm (3) 0.01 V/cm (4) 0.03 V/cm

CE0214

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Copy\Eng\03_Current Electricity\01_Eng.p65

34. Model a torch battery of length l to be made up of a thin cylindrical bar of radius 'a' and a concentric thin cylindrical shell of radius 'b' filled in between with an electrolyte of resistivity ρ (see figure). If the battery is connected to a resistance of value R, the maximum Joule heating in R will take place for:-

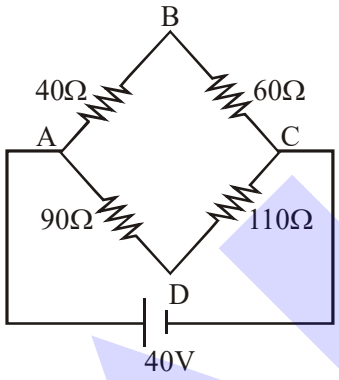
[JEE-Main-2020_Sep]



(1) $R=\frac{2\rho}{\pi l}l\,n\left(\frac{b}{a}\right)$ (2) $R=\frac{\rho}{\pi l}l\,n\left(\frac{b}{a}\right)$ (3) $R=\frac{\rho}{2\pi l}\left(\frac{b}{a}\right)$ (4) $R=\frac{\rho}{2\pi l}ln\left(\frac{b}{a}\right)$

CE0215

35.



Four resistances 40Ω, 60Ω, 90Ω and 110Ω make the arms of a quadrilateral ABCD. Across AC is a battery of emf 40V and internal resistance negligible. The potential difference across BD is V is _____.

[JEE-Main-2020_Sep]

CE0216

36. A galvanometer is used in laboratory for detecting the null point in electrical experiments. If, on passing a current of 6mA it produces a deflection of 2°, its figure of merit is close to :

[JEE-Main-2020_Sep]

(1) 3×10^{-3} A/div. (2) 333° A/div.

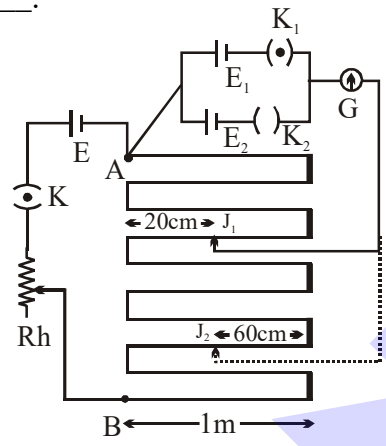
(3) 6×10^{-3} A/div. (4) 666° A/div.

CE0217

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grow\Current Elec.-Copy\Eng\03_Current Electricity\01_Eng.p65

37. In the given circuit of potentiometer, the potential difference E across AB (10m length) is larger than E_1 and E_2 as well. For key K_1 (closed), the jockey is adjusted to touch the wire at point J_1 so that there is no deflection in the galvanometer. Now the first battery (E_1) is replaced by second battery (E_2) for working by making K_1 open and K_2 closed. The galvanometer gives then null deflection at J_2. The value of $\frac{E_1}{E_2}$ is $\frac{a}{b}$, where a = ___. **[JEE-Main-2021_Feb]**

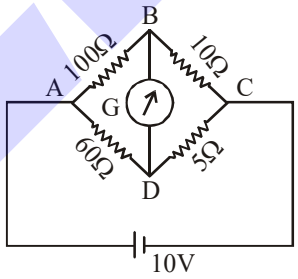
CE0321



38. The equivalent resistance of series combination of two resistors is 's'. When they are connected in parallel, the equivalent resistance is 'p'. If s = np, then the minimum value for n is ___. (Round off to the Nearest Integer) **[JEE-Main-2021_March]**

CE0322

39. The four arms of a Wheatstone bridge have resistances as shown in the figure. A galvanometer of 15Ω resistance is connected across BD. Calculate the current through the galvanometer when a potential difference of 10V is maintained across AC. **[JEE-Main-2021_March]**



(1) 2.44 μA (2) 2.44 mA (3) 4.87 mA (4) 4.87 μA

CE0323

40. A current of 5 A is passing through a non-linear magnesium wire of cross-section 0.04 m^2. At every point the direction of current density is at an angle of 60° with the unit vector of area of cross-section. The magnitude of electric field at every point of the conductor is :
(Resistivity of magnesium $\rho = 44 \times 10^{-8}$ Ωm) **[JEE-Main-2021_July]**

(1) 11×10^{-2} V/m (2) 11×10^{-7} V/m
(3) 11×10^{-5} V/m (4) 11×10^{-3} V/m

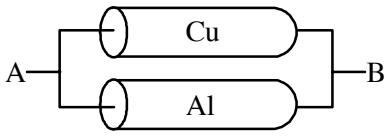
CE0324

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Capa\Eng\03_Current Electricity\01_Eng.p65

41. A Copper (Cu) rod of length 25 cm and cross-sectional area 3 mm^2 is joined with a similar Aluminium (Al) rod as shown in figure. Find the resistance of the combination between the ends A and B.
(Take Resistivity of Copper = $1.7 \times 10^{-8}\ \Omega m$ Resistivity of Aluminium = $2.6 \times 10^{-8}\ \Omega m$)

[JEE-Main-2021_July]

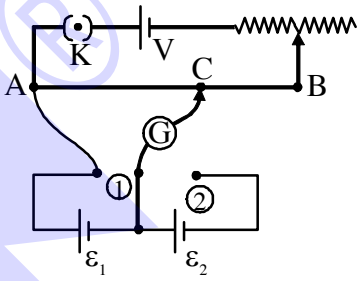
CE0325



(1) 2.170 mΩ (2) 1.420 mΩ (3) 0.0858 mΩ (4) 0.858 mΩ

42. In the given potentiometer circuit arrangement, the balancing length AC is measured to be 250 cm. When the galvanometer connection is shifted from point (1) to point (2) in the given diagram, the balancing length becomes 400 cm. The ratio of the emf of two cells, $\frac{\varepsilon_1}{\varepsilon_2}$ is :

[JEE-Main-2021_July]

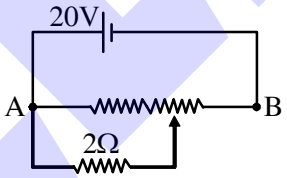


(1) $\frac{5}{3}$ (2) $\frac{8}{5}$ (3) $\frac{4}{3}$ (4) $\frac{3}{2}$

CE0326

43. The given potentiometer has its wire of resistance 10Ω. When the sliding contact is in the middle of the potentiometer wire, the potential drop across 2Ω resistor is :

[JEE-Main-2021_July]

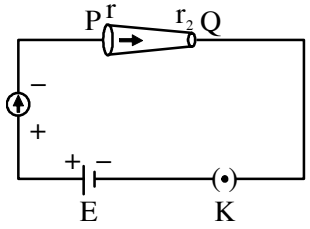


(1) 10 V (2) 5 V (3) $\frac{40}{9}$ V (4) $\frac{40}{11}$ V

CE0327

44. In the given figure, a battery of emf E is connected across a conductor PQ of length 'ℓ' and different area of cross-sections having radii r_1 and r_2 ($r_2 < r_1$).

[JEE-Main-2021_July]



Choose the correct option as one moves from P to Q :

(1) Drift velocity of electron increases. (2) Electric field decreases.

(3) Electron current decreases. (4) All of these

CE0328

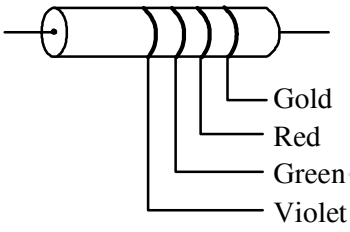
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grow\3-Current Elec.-Copa\Eng\03_Current Electricity\01_Eng.p65

45. If you are provided a set of resistances 2Ω, 4Ω, 6Ω and 8Ω. Connect these resistances so as to obtain an equivalent resistance of $\frac{46}{3}\Omega$. **[JEE-Main-2021_Aug]**

(1) 4Ω and 6Ω are in parallel with 2Ω and 8Ω in series
(2) 6Ω and 8Ω are in parallel with 2Ω and 4Ω in series
(3) 2Ω and 6Ω are in parallel with 4Ω and 8Ω in series
(4) 2Ω and 4Ω are in parallel with 6Ω and 8Ω in series

CE0329

46. The colour coding on a carbon resistor is shown in the given figure. The resistance value of the given resistor is : **[JEE-Main-2021_Aug]**



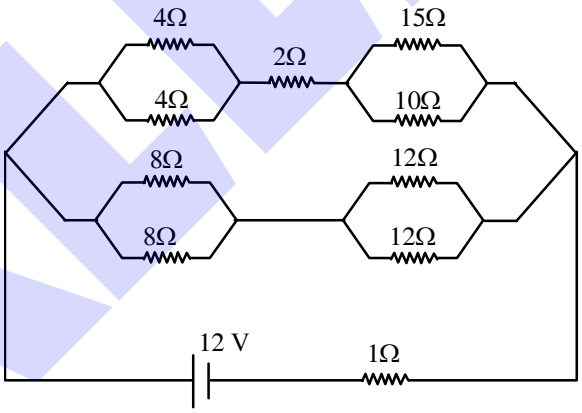
(1) (5700 ± 285) Ω (2) (7500 ± 750) Ω (3) (5700 ± 375) Ω (4) (7500 ± 375) Ω

CE0330

47. A square shaped wire with resistance of each side 3Ω is bent to form a complete circle. The resistance between two diametrically opposite points of the circle in unit of Ω will be _____. **[JEE-Main-2021_Aug]**

CE0331

48. The voltage drop across 15Ω resistance in the given figure will be_____V. **[JEE-Main-2021_Aug]**



CE0332

49. What will be the most suitable combination of three resistors A = 2Ω, B = 4Ω, C = 6Ω so that $\left(\frac{22}{3}\right)\Omega$ is equivalent resistance of combination? **[JEE-Main-2022_June]**

(A) Parallel combination of A and C connected in series with B.
(B) Parallel combination of A and B connected in series with C.
(C) Series combination of A and C connected in parallel with B.
(D) Series combination of B and C connected in parallel with A.

CE0333

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Capa\Eng\03_Current Electricity\01_Eng.p65

50. If n represents the actual number of deflections in a converted galvanometer of resistance G and shunt resistance S. Then the total current I when its figure of merit is K will be :

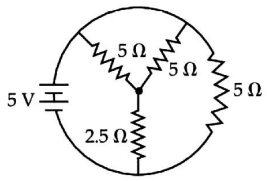
[JEE-Main-2022_June]

(A) $\frac{KS}{(S+G)}$ (B) $\frac{(G+S)}{nKS}$ (C) $\frac{nKS}{(G+S)}$ (D) $\frac{nK(G+S)}{S}$

CE0334

51. The total current supplied to the circuit as shown in figure by the 5V battery is _____A

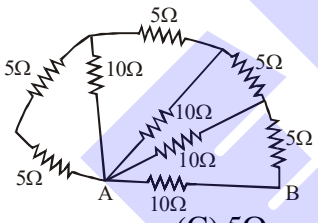
[JEE-Main-2022_June]



CE0335

52. The equivalent resistance between points A and B in the given network is :

[JEE-Main-2022_June]

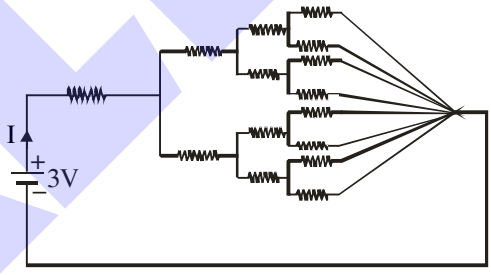


(A) 65Ω (B) 20Ω (C) 5Ω (D) 2Ω

CE0336

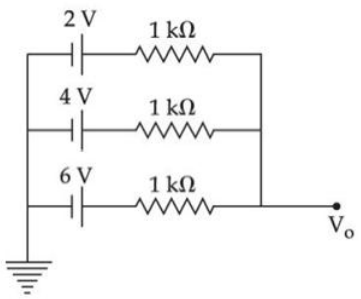
53. All resistances in figure are 1Ω each. The value of current 'I' is $\frac{a}{5}A$. The value of a is _____

[JEE-Main-2022_June]



CE0337

54. In the given figure, the value of V_0 will be ____ V. **[JEE-Main-2022_July]**

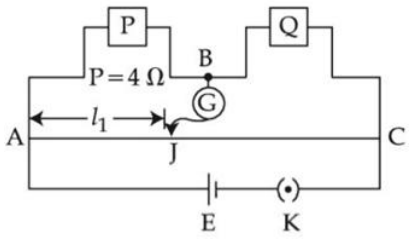


CE0338

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grow\Current Elec.-Copy\Eng\03_Current Electricity\01_Eng.p65

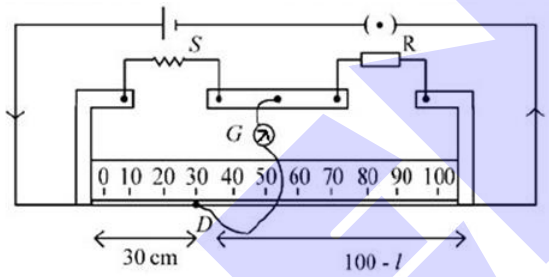
55. Resistance are connected in a meter bridge circuit as shown in the figure. The balancing length l_1 is 40cm. Now an unknown resistance x is connected in series with P and new balancing length is found to be 80cm measured from the same end. Then the value of x will be _____ Ω

[JEE-Main-2022_July]



CE0339

56. In meter bridge experiment for measuring unknown resistance 'S', the null point is obtained at a distance 30 cm from the left side as shown at point D. If R is 5.6 kΩ, then the value of unknown resistance 'S' will be _____ Ω. **[JEE-Main-2022_July]**



CE0340

57. A. The drift velocity of electrons decreases with the increase in the temperature of conductor.
B. The drift velocity is inversely proportional to the area of cross-section of given conductor.
C. The drift velocity does not depend on the applied potential difference to the conductor.
D. The drift velocity of electron is inversely proportional to the length of the conductor.
E. The drift velocity increases with the increase in the temperature of conductor.
Choose the correct answer from the options given below: **[JEE-Main-2022_July]**
(A) A and B only (B) A and D only (C) B and E only (D) B and C only

CE0341

58. Two metallic wires of identical dimensions are connected is series. If σ_1 and σ_2 are the conductivities of the these wires respectively, the effective conductivity of the combination is :

[JEE-Main-2022_July]

(A) $\frac{\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$ (B) $\frac{2\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$ (C) $\frac{\sigma_1 + \sigma_2}{2\sigma_1 \sigma_2}$ (D) $\frac{\sigma_1 + \sigma_2}{\sigma_1 \sigma_2}$

CE0342

59. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A: Alloys such as constantan and manganin are used in making standard resistance coils.
Reason R: Constantan and manganin have very small value of temperature coefficient of resistance.
In the light of the above statements, choose the correct answer from the options given below.

[JEE-Main-2022_July]

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is NOT the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

CE0343

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Capa\Eng\03_Current Electricity\01_Eng.p65

EXERCISE - JA

1. Heater of an electric kettle is made of a wire of length L and diameter d. It takes 4 minutes to raise the temperature of 0.5 kg water by 40 K. This heater is replaced by a new heater having two wires of the same material, each of length L and diameter 2d. The way these wires are connected is given in the options. How much time in minutes will it take to raise the temperature of the same amount of water by 40 K ? **[JEE Advanced 2014]**

(A) 4 if wires are in parallel

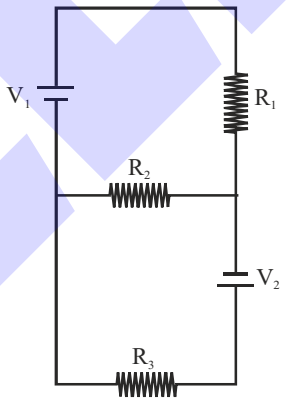
(B) 2 if wires are in series

(C) 1 if wires are in series

(D) 0.5 if wires are in parallel

CE0159

2. Two ideal batteries of emf V_1 and V_2 and three resistances R_1, R_2 and R_3 are connected as shown in the figure. The current in resistance R_2 would be zero if :- **[JEE Advanced 2014]**



(A) $V_1 = V_2$ and $R_1 = R_2 = R_3$

(B) $V_1 = V_2$ and $R_1 = 2R_2 = R_3$

(C) $V_1 = 2V_2$ and $2R_1 = 2R_2 = R_3$

(D) $2V_1 = V_2$ and $2R_1 = R_2 = R_3$

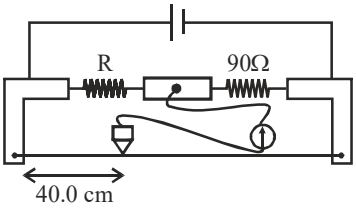
CE0160

3. A galvanometer gives full scale deflection with 0.006 A current. By connecting it to a 4990 Ω resistance, it can be converted into a voltmeter of range 0 - 30 V. If connected to a $\frac{2n}{249}\Omega$ resistance, it becomes an ammeter of range 0 - 1.5 A. The value of n is :- **[JEE Advanced 2014]**

CE0161

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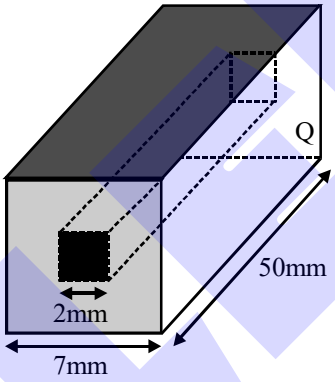
4. During an experiment with a metre bridge, the galvanometer shows a null point when the jockey is pressed at 40.0 cm using a standard resistance of 90Ω, as shown in the figure. The least count of the scale used in the metre bridge is 1mm. The unknown resistance is :- **[JEE Advanced 2014]**



(A) 60 ± 0.15 Ω (B) 135 ± 0.56 Ω (C) 60 ± 0.25 Ω (D) 135 ± 0.23 Ω

CE0162

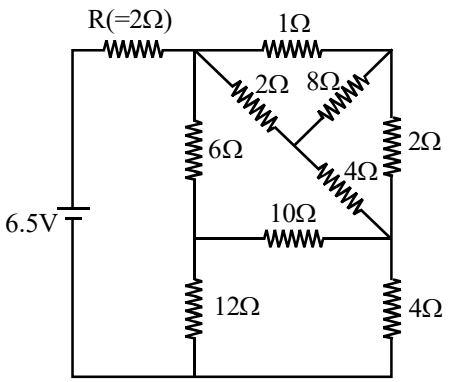
5. In an aluminum (Al) bar of square cross section, a square hole is drilled and is filled with iron (Fe) as shown in the figure. The electrical resistivities of Al and Fe are $2.7\times10^{-8}\ \Omega$ m and $1.0\times10^{-7}\ \Omega$m, respectively. The electrical resistance between the two faces P and Q of the composite bar is : **[JEE Advanced-2015]**



(A) $\frac{2475}{64}\mu\Omega$ (B) $\frac{1875}{64}\mu\Omega$ (C) $\frac{1875}{49}\mu\Omega$ (D) $\frac{2475}{132}\mu\Omega$

CE0163

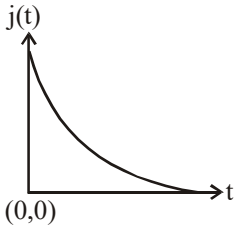
6. In the following circuit, the current through the resistor R (= 2Ω) is I Amperes. The value of I is **[JEE Advanced-2015]**

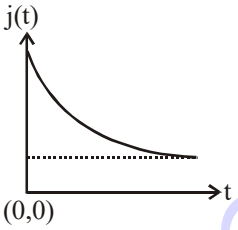


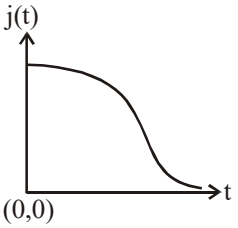
CE0164

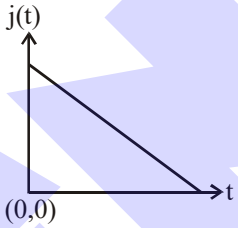
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\-Current Elec.-Capa\Eng\03_Current Electricity\01_Eng.p65

7. An infinite line charge of uniform electric charge density λ lies along the axis of an electrically conducting infinite cylindrical shell of radius R. At time t = 0, the space inside the cylinder is filled with a material of permittivity ε and electrical conductivity σ. The electrical conduction in the material follows Ohm's law. Which one of the following graphs best describes the subsequent variation of the magnitude of current density j(t) at any point in the material? **[JEE Advanced-2016]**

(A) 

(B) 

(C) 

(D) 

CE0165

8. An incandescent bulb has a thin filament of tungsten that is heated to high temperature by passing an electric current. The hot filament emits black-body radiation. The filament is observed to break up at random locations after a sufficiently long time of operation due to non-uniform evaporation of tungsten from the filament. If the bulb is powered at constant voltage, which of the following statement(s) is(are) true? **[JEE Advanced-2016]**

(A) The temperature distribution over the filament is uniform

(B) The resistance over small sections of the filament decreases with time

(C) The filament emits more light at higher band of frequencies before it breaks up

(D) The filament consumes less electrical power towards the end of the life of the bulb

CE0166

9. Consider two identical galvanometers and two identical resistors with resistance R. If the internal resistance of the galvanometers $R_C < R/2$, which of the following statement(s) about any one of the galvanometers is(are) true ? **[JEE Advanced-2016]**

(A) The maximum voltage range is obtained when all the components are connected in series

(B) The maximum voltage range is obtained when the two resistors and one galvanometer are connected in series, and the second galvanometer is connected in parallel to the first galvanometer

(C) The maximum current range is obtained when all the components are connected in parallel

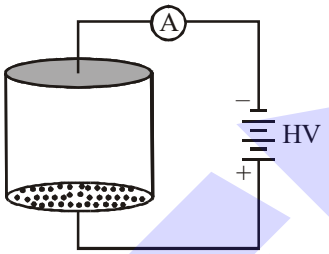
(D) The maximum current range is obtained when the two galvanometers are connected in series and the combination is connected in parallel with both the resistors.

CE0167

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav\3-Current Elec.-Capa\Eng\03_Current Electricity\01_Eng.p65

Paragraph for Questions No. 10 and 11

Consider an evacuated cylindrical chamber of height h having rigid conducting plates at the ends and an insulating curved surface as shown in the figure. A number of spherical balls made of a light weight and soft material and coated with a conducting material are placed on the bottom plate. The balls have a radius r <<h. Now a high voltage source (HV) is connected across the conducting plates such that the bottom plate is at $+V_0$ and the top plate at $-V_0$. Due to their conducting surface, the balls will get charged, will become equipotential with the plate and are repelled by it. The balls will eventually collide with the top plate, where the coefficient of restitution can be taken to be zero due to the soft nature of the material of the balls. The electric field in the chamber can be considered to be that of a parallel plate capacitor. Assume that there are no collision between the balls and the interaction between them is negligible. (Ignore gravity) **[JEE Advanced-2016]**



10. Which of the following statements is correct ?

(A) The balls will bounce back to the bottom plate carrying the opposite charge they went up with

(B) the balls will execute simple harmonic motion between the two plates

(C) The balls will bounce back to the bottom plate carrying the same charge they went up with

(D) The balls will stick to the top plate and remain there

CE0168

11. The average current in the steady state registered by the ammeter in the circuit will be :

(A) Proportional to $V_0^{1/2}$ (B) Proportional to V_0^2

(C) Proportional to the potential V_0 (D) Zero

CE0168

12. Two identical moving coil galvanometer have 10 Ω resistance and full scale deflection at 2 μA current. One of them is converted into a voltmeter of 100 mV full scale reading and the other into an Ammeter of 1 mA full scale current using appropriate resistors. These are then used to measure the voltage and current in the Ohm's law experiment with R = 1000 Ω resistor by using an ideal cell. Which of the following statement(s) is/are correct ? **[JEE Advanced-2019]**

(1) The measured value of R will be 978 Ω < R < 982 Ω.

(2) The resistance of the Voltmeter will be 100 kΩ.

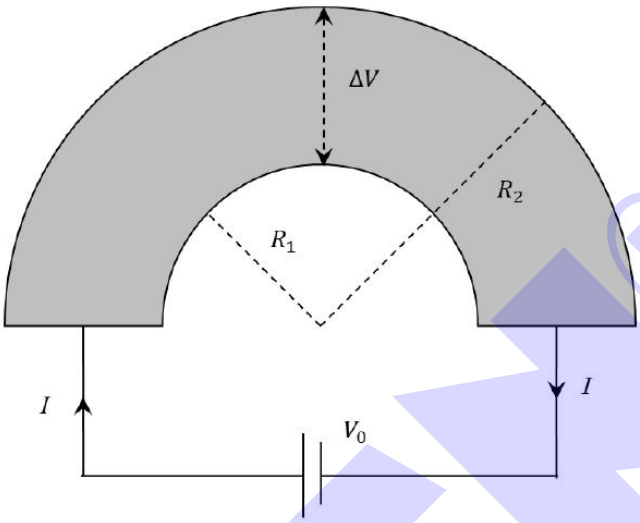
(3) The resistance of the Ammeter will be 0.02 Ω (round off to 2nd decimal place)

(4) If the ideal cell is replaced by a cell having internal resistance of 5 Ω then the measured value of R will be more than 1000 Ω.

CE0169

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13. Shown in the figure is a semicircular metallic strip that has thickness t and resistivity ρ. Its inner radius is R_1 and outer radius is R_2. If a voltage V_0 is applied between its two ends, a current I flows in it. In addition, it is observed that a transverse voltage ΔV develops between its inner and outer surfaces due to purely kinetic effects of moving electrons (ignore any role of the magnetic field due to the current). Then (figure is schematic and not drawn to scale)- **[JEE Advanced-2020]**



(A) $I = \frac{V_0 t}{\pi\rho} \ln\left(\frac{R_2}{R_1}\right)$

(B) the outer surface is at a higher voltage than the inner surface

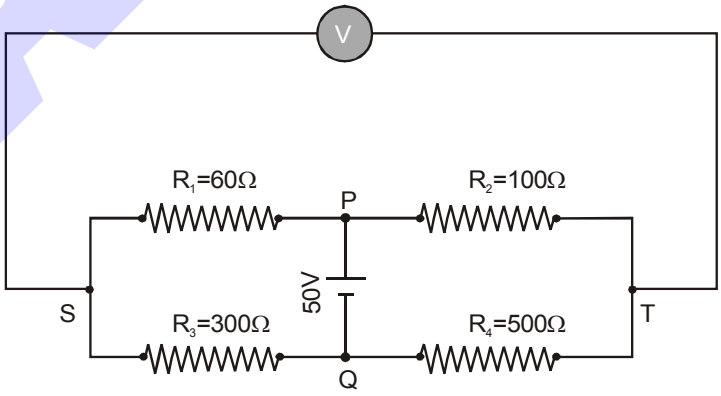
(C) the outer surface is at a lower voltage than the inner surface

(D) $\Delta V \propto I^2$

CE0218

14. In the balanced condition, the values of the resistances of the four arms of a Wheatstone bridge are shown in the figure below. The resistance R_3 has temperature coefficient 0.0004 $°C^{-1}$. If the temperature of R_3 is increased by 100°C, the voltage developed between S and T will be _____ volt.

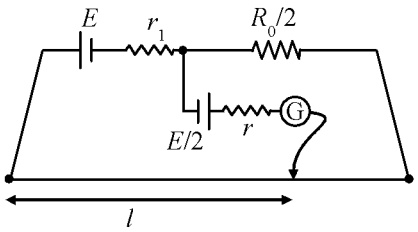
[JEE Advanced-2020]



CE0219

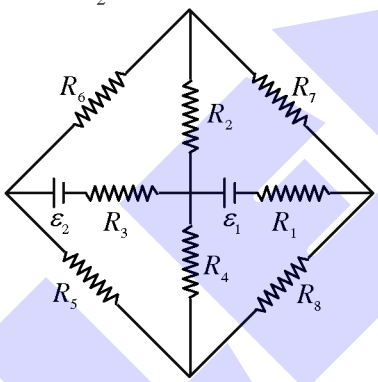
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15. In order to measure the internal resistance r_1 of a cell of emf E, a meter bridge of wire resistance $R_0 = 50\Omega$, a resistance $R_0/2$, another cell of emf E/2 (internal resistance r) and a galvanometer G are used in a circuit, as shown in the figure. If the null point is found at $\ell = 72$ cm, then the value of r_1 =_____ Ω. **[JEE Advanced-2021]**



CE0220

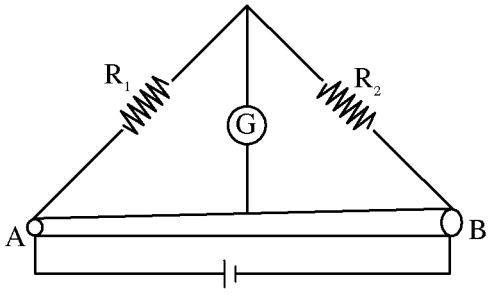
16. The figure shows a circuit having eight resistances of 1Ω each, labelled R_1 to R_8, and two ideal batteries with voltages $\varepsilon_1 = 12$ V and $\varepsilon_2 = 6$ V. **[JEE Advanced-2022]**



Which of the following statement(s) is(are) correct?
(A) The magnitude of current flowing through R_1 is 7.2 A.
(B) The magnitude of current flowing through R_2 is 1.2 A.
(C) The magnitude of current flowing through R_3 is 4.8 A.
(D) The magnitude of current flowing through R_5 is 2.4 A.

CE0344

17. Two resistances $R_1 = X\Omega$ and $R_2 = 1\Omega$ are connected to a wire AB of uniform resistivity, as shown in the figure. The radius of the wire varies linearly along its axis from 0.2 mm at A to 1 mm at B. A galvanometer (G) connected to the center of the wire, 50 cm from each end along its axis, shows zero deflection when A and B are connected to a battery. The value of X is _____. **[JEE Advanced-2022]**

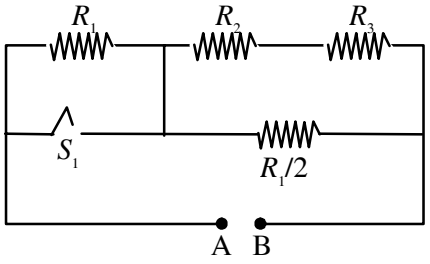


CE0345

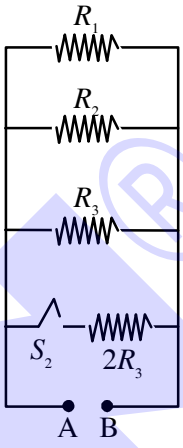
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18. In Circuit-1 and Circuit-2 shown in the figures, $R_1 = 1\Omega$, $R_2 = 2\Omega$ and $R_3 = 3\Omega$. P_1 and P_2 are the power dissipations in Circuit-1 and Circuit-2 when the switches S_1 and S_2 are in open conditions, respectively.

Q_1 and Q_2 are the power dissipations in Circuit-1 and Circuit-2 when the switches S_1 and S_2 are in closed conditions, respectively.



Circuit-1



Circuit-2

Which of the following statement(s) is(are) correct? **[JEE Advanced-2022]**

(A) When a voltage source of 6 V is connected across A and B in both circuits, $P_1 < P_2$.

(B) When a constant current source of 2 Amp is connected across A and B in both circuits, $P_1 > P_2$.

(C) When a voltage source of 6 V is connected across A and B in Circuit-1, $Q_1 > P_1$.

(D) When a constant current source of 2 Amp is connected across A and B in both circuits, $Q_2 < Q_1$

CE0346

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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	D	C	C	B	B	B	D	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	A	C	A	A	A	C	A	C	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	A	C	B	C	C	B	D	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	C	B	A	B	A	A	D	D	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	D	C	C	C	D	B	C	A	B	B
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	A	B	B	C	A	B	D	C	C
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	D	B	D	A	B	B	C	A	D	B

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	A	B	C	B	D	A	C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	B	A	B	C	B	A	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	B	D	A	A	B	C	A	C	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	A	D	B	B	B	D	B	C	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	D	C	C	C	B	A	D	B	A
Que.	51	52	53	54	55	56	57			
Ans.	A,B,C,D	A,D	A,B,C	A,B,C,D	A,C	B,C	A,B,C			
Que.	58	A	B	C	D	59	A	B	C	
Ans.		P	Q	R	P		Q	P	P	

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EXERCISE # S

1. $I\rho/A$ **2.** $S = e\,n\,l\,\langle v\rangle / j$ **3.** $p = I\,m_e\,l/e$ **4.** 600Ω **5.** 7.5 mA **6.** 1 V

7. I = 2.5 A, V = 3.5 Volts **8.** 20/3 V **9.** $x = \frac{4}{3}$ V, $12\frac{1}{3}$ V, $\frac{1}{15}$ A **10.** 5

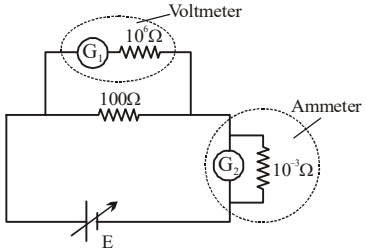
11. (a) (i) in series, (ii) all in parallel: n^2.
(b) (i) Join 1Ω, 2Ω in parallel and the combination in series with 3Ω,
(ii) parallel combination of 2 Ω and 3 Ω in series with 1 Ω,; (iii) all in series, (iv) all in parallel.
(c) (i) (16/3) Ω, (ii) 5 R.

12. $\frac{8}{7}$R **13.** 9Ω **14.** 3.5 A **16.** 12A, –20W **17.** 5 **18.** $\sqrt{R_1R_2}$

19. 25 **20.** 3 **21.** 4/9 kg/sec., 450 sec **22.** 4 **23.** 20 ohm

24. $R_1 = 0.0278\Omega$, $R_2 = 0.25\Omega$, $R_3 = 2.5\Omega$ **25.** 233.3Ω, 144V

26.



27. Battery should be connected across A and B. Out put can be taken across the terminals A and C or B and C

28. This is true for $r_1 = r_2$; So R_2 given most accurate value **29.** (a) +ve, $E_\ell > E$ (b) –ve

30. 2.25 V **31.** 4 ohm **32.** 46.67 cm

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	1	2	1	2	2	3	4	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	1	2	1	2	4	3	2	2	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	4	2	4	1	2	4	1	4	3	2
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	4	30	1	4	2	1	1	4	3	3
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	4	1	3	1	4	4	3	6	B	D
Que.	51	52	53	54	55	56	57	58	59	
Ans.	2	C	8	4	20	2400	B	B	A	

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,D	A,B,D	5	C	B	1	A	C,D	B,C	A
Que.	11	12	13	14	15	16	17	18		
Ans.	B	1,3	A,C,D	0.26 to 0.28	3	A,B,C,D	5	A,B,C		

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\2-Electro-Grow\3-Current Elec.-Copy\Eng\03_Current Electricity\01_Eng.p65